Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.39)

Metric

$$ds^2 = -u_4^2 \mathrm{d}t \otimes \mathrm{d}t + (-\frac{1}{C^2 r^2 - 1})\mathrm{d}r \otimes \mathrm{d}r + r^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + r^2 \sin(\theta)^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Coordinates

$$u_4(t, r, \theta, \phi) = u_4$$

Null Tetrad

Covariant components

$$l = (\sqrt{\frac{1}{2}} u_4)\, dt + (\sqrt{\frac{1}{2}} \sqrt{-\frac{1}{C^2 r^2 - 1}})\, dr$$

$$n = (\sqrt{\frac{1}{2}} u_4)\, dt + (-\sqrt{\frac{1}{2}} \sqrt{-\frac{1}{C^2 r^2 - 1}})\, dr$$

$$m = (\frac{1}{2} \sqrt{2} r)\, d\theta + (\frac{1}{2} i \sqrt{2} r \sin(\theta))\, d\phi$$

$$\overline{m} = (\frac{1}{2} \sqrt{2} r)\, d\theta + (-\frac{1}{2} i \sqrt{2} r \sin(\theta))\, d\phi$$

Contravariant components

$$l = (-\frac{\sqrt{2}}{2\,u_4})\,dt + (-\frac{\sqrt{2}C^2r^2}{2\,\sqrt{Cr+1}\sqrt{|Cr-1|}} + \frac{\sqrt{2}}{2\,\sqrt{Cr+1}\sqrt{|Cr-1|}})\,dr$$

$$n = (-\frac{\sqrt{2}}{2\,u_4})\,dt + (\frac{\sqrt{2}C^2r^2}{2\,\sqrt{Cr+1}\sqrt{|Cr-1|}} - \frac{\sqrt{2}}{2\,\sqrt{Cr+1}\sqrt{|Cr-1|}})\,dr$$

$$m = (\frac{\sqrt{2}}{2\,r})\,d\theta + (\frac{i\,\sqrt{2}}{2\,r\,\sin(\theta)})\,d\phi$$

$$\overline{m} = (\frac{\sqrt{2}}{2\,r})\,d\theta + (-\frac{i\,\sqrt{2}}{2\,r\,\sin(\theta)})\,d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\,u_4}\,\partial_t - \frac{\sqrt{2}C^2r^2}{2\,\sqrt{Cr+1}\sqrt{|Cr-1|}} + \frac{\sqrt{2}}{2\,\sqrt{Cr+1}\sqrt{|Cr-1|}}\,\partial_r$$

$$\Delta = -\frac{\sqrt{2}}{2\,u_4}\,\partial_t + \frac{\sqrt{2}C^2r^2}{2\,\sqrt{Cr+1}\sqrt{|Cr-1|}} - \frac{\sqrt{2}}{2\,\sqrt{Cr+1}\sqrt{|Cr-1|}}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2\,r}\,\partial_\theta + \frac{i\,\sqrt{2}}{2\,r\,\sin(\theta)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,r}\,\partial_\theta - \frac{i\,\sqrt{2}}{2\,r\,\sin(\theta)}\,\partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}C^2 r}{2\sqrt{Cr+1}\sqrt{|Cr-1|}} - \frac{\sqrt{2}}{2\sqrt{Cr+1}r\sqrt{|Cr-1|}}$$

$$\sigma = 0$$

$$\kappa = -\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,ru_4} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r\sin(\theta)u_4}$$

$$\mu = \frac{\sqrt{2}C^2 r}{2\sqrt{Cr+1}\sqrt{|Cr-1|}} - \frac{\sqrt{2}}{2\sqrt{Cr+1}r\sqrt{|Cr-1|}}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,ru_4} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r\sin(\theta)u_4}$$

$$\alpha = -\frac{\sqrt{2}\cos(\theta)}{4\,r\sin(\theta)}$$

$$\beta = \frac{\sqrt{2}\cos(\theta)}{4\,r\sin(\theta)}$$

$$\tau = -\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,ru_4} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r\sin(\theta)u_4}$$

$$\pi = \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,ru_4} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r\sin(\theta)u_4}$$

$$\epsilon = \frac{\sqrt{Cr+1}C^3r^2\sqrt{|Cr-1|}}{4\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})} - \frac{\sqrt{Cr+1}C^2r^2|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4}{2\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})u_4} + \frac{\sqrt{Cr+1}C^2r|Cr-1|^{\frac{3}{2}}}{4\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})}$$
$$- \frac{\sqrt{Cr+1}C^2r\sqrt{|Cr-1|}}{4\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})} + \frac{\sqrt{Cr+1}|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4}{2\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})u_4}$$

$$\gamma = \frac{\sqrt{Cr+1}C^3r^2\sqrt{|Cr-1|}}{4\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})} - \frac{\sqrt{Cr+1}C^2r^2|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4}{2\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})u_4} + \frac{\sqrt{Cr+1}C^2r|Cr-1|^{\frac{3}{2}}}{4\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})}$$
$$- \frac{\sqrt{Cr+1}C^2r\sqrt{|Cr-1|}}{4\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})} + \frac{\sqrt{Cr+1}|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4}{2\,(\sqrt{2}C^3r^3-\sqrt{2}C^2r^2-\sqrt{2}Cr+\sqrt{2})u_4}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{C^3r^3\frac{\partial^2}{(\partial r)^2}u_4}{8\,|Cr-1|u_4} + \frac{C^3r^2\frac{\partial}{\partial r}u_4}{8\,|Cr-1|u_4} - \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{8\,u_4} - \frac{C^3r}{8\,|Cr-1|} - \frac{3\,C^2r\frac{\partial}{\partial r}u_4}{8\,u_4} + \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{8\,|Cr-1|u_4} + \frac{3}{8}\,C^2 - \frac{C^2r\frac{\partial}{\partial r}u_4}{8\,|Cr-1|u_4} + \frac{C^2}{8\,|Cr-1|}$$
$$+ \frac{Cr\frac{\partial^2}{(\partial r)^2}u_4}{8\,|Cr-1|u_4} - \frac{C\frac{\partial}{\partial r}u_4}{4\,|Cr-1|u_4} + \frac{\frac{\partial^2}{(\partial r)^2}u_4}{8\,u_4} + \frac{\frac{\partial}{\partial r}u_4}{4\,ru_4} - \frac{\frac{\partial^2}{(\partial r)^2}u_4}{8\,|Cr-1|u_4} - \frac{C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r|Cr-1|\sin(\theta)u_4} - \frac{C\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r|Cr-1|u_4} + \frac{\frac{\partial}{\partial r}u_4}{4\,r|Cr-1|u_4}$$
$$+ \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2u_4} + \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2|Cr-1|u_4} - \frac{C\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r|Cr-1|\sin(\theta)^2u_4} + \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2\sin(\theta)^2u_4} + \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2|Cr-1|\sin(\theta)^2u_4}$$

$$\Phi_{01} = \frac{\sqrt{Cr+1}C\frac{\partial^2}{\partial r\partial\theta}u_4}{4\sqrt{|Cr-1|}u_4} + \frac{i\sqrt{Cr+1}C\frac{\partial^2}{\partial r\partial\phi}u_4}{4\sqrt{|Cr-1|}\sin(\theta)u_4} - \frac{\sqrt{Cr+1}C\frac{\partial}{\partial\theta}u_4}{4\,r\sqrt{|Cr-1|}u_4} - \frac{\sqrt{Cr+1}\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,r\sqrt{|Cr-1|}u_4}$$
$$- \frac{i\sqrt{Cr+1}C\frac{\partial}{\partial\phi}u_4}{4\,r\sqrt{|Cr-1|}\sin(\theta)u_4} - \frac{i\sqrt{Cr+1}\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,r\sqrt{|Cr-1|}\sin(\theta)u_4} + \frac{\sqrt{Cr+1}\frac{\partial}{\partial\theta}u_4}{4\,r^2\sqrt{|Cr-1|}u_4} + \frac{i\sqrt{Cr+1}\frac{\partial}{\partial\phi}u_4}{4\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}$$

$$\Phi_{02} = \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{4\,r^2\sin(\theta)u_4} - \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{4\,r^2u_4} - \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}u_4}{2\,r^2\sin(\theta)u_4} + \frac{i\,\cos(\theta)\frac{\partial}{\partial\phi}u_4}{2\,r^2\sin(\theta)^2u_4} + \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{4\,r^2\sin(\theta)^2u_4}$$

$$\Phi_{10} = \frac{\sqrt{Cr+1}C\frac{\partial^2}{\partial r\partial\theta}u_4}{4\sqrt{|Cr-1|}u_4} - \frac{i\sqrt{Cr+1}C\frac{\partial^2}{\partial r\partial\phi}u_4}{4\sqrt{|Cr-1|}\sin(\theta)u_4} - \frac{\sqrt{Cr+1}C\frac{\partial}{\partial\theta}u_4}{4\,r\sqrt{|Cr-1|}u_4} - \frac{\sqrt{Cr+1}\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,r\sqrt{|Cr-1|}u_4}$$
$$+ \frac{i\sqrt{Cr+1}C\frac{\partial}{\partial\phi}u_4}{4\,r\sqrt{|Cr-1|}\sin(\theta)u_4} + \frac{i\sqrt{Cr+1}\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,r\sqrt{|Cr-1|}\sin(\theta)u_4} + \frac{\sqrt{Cr+1}\frac{\partial}{\partial\theta}u_4}{4\,r^2\sqrt{|Cr-1|}u_4} - \frac{i\sqrt{Cr+1}\frac{\partial}{\partial\phi}u_4}{4\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}$$

$$\Phi_{11} = \frac{C^3 r^3 \frac{\partial^2}{(\partial r)^2} u_4}{16\,|Cr-1| u_4} - \frac{C^3 r^2 \frac{\partial}{\partial r} u_4}{16\,|Cr-1| u_4} - \frac{3\,C^2 r^2 \frac{\partial^2}{(\partial r)^2} u_4}{16\,u_4} + \frac{C^3 r}{16\,|Cr-1|} - \frac{5\,C^2 r \frac{\partial}{\partial r} u_4}{16\,u_4} - \frac{C^2 r^2 \frac{\partial^2}{(\partial r)^2} u_4}{16\,|Cr-1| u_4} + \frac{5}{16}\,C^2 + \frac{C^2 r \frac{\partial}{\partial r} u_4}{16\,|Cr-1| u_4} - \frac{C^2}{16\,|Cr-1|}$$
$$- \frac{Cr \frac{\partial^2}{(\partial r)^2} u_4}{16\,|Cr-1| u_4} + \frac{C \frac{\partial}{\partial r} u_4}{8\,|Cr-1| u_4} + \frac{3 \frac{\partial^2}{(\partial r)^2} u_4}{16\,u_4} + \frac{\frac{\partial}{\partial r} u_4}{8\,r u_4} + \frac{\frac{\partial^2}{(\partial r)^2} u_4}{16\,|Cr-1| u_4} + \frac{C \cos(\theta) \frac{\partial}{\partial \theta} u_4}{16\,r|Cr-1| \sin(\theta) u_4} + \frac{C \frac{\partial^2}{(\partial \theta)^2} u_4}{16\,r|Cr-1| u_4} - \frac{\frac{\partial}{\partial r} u_4}{8\,r|Cr-1| u_4}$$
$$+ \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{16\,r^2 \sin(\theta) u_4} + \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{16\,r^2 u_4} - \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{16\,r^2|Cr-1| \sin(\theta) u_4} - \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{16\,r^2|Cr-1| u_4} + \frac{C \frac{\partial^2}{(\partial \phi)^2} u_4}{16\,r|Cr-1| \sin(\theta)^2 u_4} + \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{16\,r^2 \sin(\theta)^2 u_4} - \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{16\,r^2|Cr-1| \sin(\theta)^2 u_4}$$

$$\Phi_{12} = -\frac{\sqrt{Cr+1} C \frac{\partial^2}{\partial r \partial \theta} u_4}{4\,\sqrt{|Cr-1|} u_4} - \frac{i\,\sqrt{Cr+1} C \frac{\partial^2}{\partial r \partial \phi} u_4}{4\,\sqrt{|Cr-1|} \sin(\theta) u_4} + \frac{\sqrt{Cr+1} C \frac{\partial}{\partial \theta} u_4}{4\,r\sqrt{|Cr-1|} u_4} + \frac{\sqrt{Cr+1} \frac{\partial^2}{\partial r \partial \theta} u_4}{4\,r\sqrt{|Cr-1|} u_4}$$
$$+ \frac{i\,\sqrt{Cr+1} C \frac{\partial}{\partial \phi} u_4}{4\,r\sqrt{|Cr-1|} \sin(\theta) u_4} + \frac{i\,\sqrt{Cr+1} \frac{\partial^2}{\partial r \partial \phi} u_4}{4\,r\sqrt{|Cr-1|} \sin(\theta) u_4} - \frac{\sqrt{Cr+1} \frac{\partial}{\partial \theta} u_4}{4\,r^2\sqrt{|Cr-1|} u_4} - \frac{i\,\sqrt{Cr+1} \frac{\partial}{\partial \phi} u_4}{4\,r^2\sqrt{|Cr-1|} \sin(\theta) u_4}$$

$$\Phi_{20} = \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{4\,r^2 \sin(\theta) u_4} - \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{4\,r^2 u_4} + \frac{i\,\frac{\partial^2}{\partial \theta \partial \phi} u_4}{2\,r^2 \sin(\theta) u_4} - \frac{i\,\cos(\theta) \frac{\partial}{\partial \phi} u_4}{2\,r^2 \sin(\theta)^2 u_4} + \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{4\,r^2 \sin(\theta)^2 u_4}$$

$$\Phi_{21} = -\frac{\sqrt{Cr+1} C \frac{\partial^2}{\partial r \partial \theta} u_4}{4\,\sqrt{|Cr-1|} u_4} + \frac{i\,\sqrt{Cr+1} C \frac{\partial^2}{\partial r \partial \phi} u_4}{4\,\sqrt{|Cr-1|} \sin(\theta) u_4} + \frac{\sqrt{Cr+1} C \frac{\partial}{\partial \theta} u_4}{4\,r\sqrt{|Cr-1|} u_4} + \frac{\sqrt{Cr+1} \frac{\partial^2}{\partial r \partial \theta} u_4}{4\,r\sqrt{|Cr-1|} u_4}$$
$$- \frac{i\,\sqrt{Cr+1} C \frac{\partial}{\partial \phi} u_4}{4\,r\sqrt{|Cr-1|} \sin(\theta) u_4} - \frac{i\,\sqrt{Cr+1} \frac{\partial^2}{\partial r \partial \phi} u_4}{4\,r\sqrt{|Cr-1|} \sin(\theta) u_4} - \frac{\sqrt{Cr+1} \frac{\partial}{\partial \theta} u_4}{4\,r^2\sqrt{|Cr-1|} u_4} + \frac{i\,\sqrt{Cr+1} \frac{\partial}{\partial \phi} u_4}{4\,r^2\sqrt{|Cr-1|} \sin(\theta) u_4}$$

$$\Phi_{22} = -\frac{C^3 r^3 \frac{\partial^2}{(\partial r)^2} u_4}{8\,|Cr-1| u_4} + \frac{C^3 r^2 \frac{\partial}{\partial r} u_4}{8\,|Cr-1| u_4} - \frac{C^2 r^2 \frac{\partial^2}{(\partial r)^2} u_4}{8\,u_4} - \frac{C^3 r}{8\,|Cr-1|} - \frac{3\,C^2 r \frac{\partial}{\partial r} u_4}{8\,u_4} + \frac{C^2 r^2 \frac{\partial^2}{(\partial r)^2} u_4}{8\,|Cr-1| u_4} + \frac{3}{8}\,C^2 - \frac{C^2 r \frac{\partial}{\partial r} u_4}{8\,|Cr-1| u_4} + \frac{C^2}{8\,|Cr-1|}$$
$$+ \frac{Cr \frac{\partial^2}{(\partial r)^2} u_4}{8\,|Cr-1| u_4} - \frac{C \frac{\partial}{\partial r} u_4}{4\,|Cr-1| u_4} + \frac{\frac{\partial^2}{(\partial r)^2} u_4}{8\,u_4} + \frac{\frac{\partial}{\partial r} u_4}{4\,r u_4} - \frac{\frac{\partial^2}{(\partial r)^2} u_4}{8\,|Cr-1| u_4} - \frac{C \cos(\theta) \frac{\partial}{\partial \theta} u_4}{8\,r|Cr-1| \sin(\theta) u_4} - \frac{C \frac{\partial^2}{(\partial \theta)^2} u_4}{8\,r|Cr-1| u_4} + \frac{\frac{\partial}{\partial r} u_4}{4\,r|Cr-1| u_4}$$
$$+ \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{8\,r^2 \sin(\theta) u_4} + \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{8\,r^2 u_4} + \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{8\,r^2|Cr-1| \sin(\theta) u_4} + \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{8\,r^2|Cr-1| u_4} - \frac{C \frac{\partial^2}{(\partial \phi)^2} u_4}{8\,r|Cr-1| \sin(\theta)^2 u_4} + \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{8\,r^2 \sin(\theta)^2 u_4} + \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{8\,r^2|Cr-1| \sin(\theta)^2 u_4}$$

$$\Lambda = \frac{C^2 r^2 \frac{\partial^2}{(\partial r)^2} u_4}{12\,u_4} + \frac{C^2 r \frac{\partial}{\partial r} u_4}{4\,u_4} + \frac{1}{4}\,C^2 - \frac{\frac{\partial^2}{(\partial r)^2} u_4}{12\,u_4} - \frac{\frac{\partial}{\partial r} u_4}{6\,r u_4} - \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{12\,r^2 \sin(\theta) u_4} - \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{12\,r^2 u_4} - \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{12\,r^2 \sin(\theta)^2 u_4}$$

Weyl Tensor Components

$$\Psi_0 = \frac{C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r|Cr-1|\sin(\theta)u_4} - \frac{C\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r|Cr-1|u_4} - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2u_4} - \frac{i\,C\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)u_4} + \frac{i\,C\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)^2u_4} - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2|Cr-1|u_4}$$
$$+ \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r^2\sin(\theta)u_4} - \frac{i\,\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r^2\sin(\theta)^2u_4} + \frac{C\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r|Cr-1|\sin(\theta)^2u_4} + \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r^2|Cr-1|\sin(\theta)u_4} - \frac{i\,\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r^2|Cr-1|\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2|Cr-1|\sin(\theta)^2u_4}$$

$$\Psi_1 = \frac{\sqrt{Cr+1}C\frac{\partial^2}{\partial r\partial\theta}u_4}{4\sqrt{|Cr-1|}u_4} + \frac{i\sqrt{Cr+1}C\frac{\partial^2}{\partial r\partial\phi}u_4}{4\sqrt{|Cr-1|}\sin(\theta)u_4} - \frac{\sqrt{Cr+1}C\frac{\partial}{\partial\theta}u_4}{4\,r\sqrt{|Cr-1|}u_4} - \frac{\sqrt{Cr+1}\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,r\sqrt{|Cr-1|}u_4}$$
$$- \frac{i\sqrt{Cr+1}C\frac{\partial}{\partial\phi}u_4}{4\,r\sqrt{|Cr-1|}\sin(\theta)u_4} - \frac{i\sqrt{Cr+1}\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,r\sqrt{|Cr-1|}\sin(\theta)u_4} + \frac{\sqrt{Cr+1}\frac{\partial}{\partial\theta}u_4}{4\,r^2\sqrt{|Cr-1|}u_4} + \frac{i\sqrt{Cr+1}\frac{\partial}{\partial\phi}u_4}{4\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}$$

$$\Psi_2 = \frac{C^3r^3\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} - \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{12\,u_4} - \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} - \frac{Cr\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} + \frac{C\frac{\partial}{\partial r}u_4}{12\,|Cr-1|u_4} + \frac{\frac{\partial^2}{(\partial r)^2}u_4}{12\,u_4} - \frac{\frac{\partial}{\partial r}u_4}{12\,ru_4} + \frac{\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} + \frac{C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{24\,r|Cr-1|\sin(\theta)u_4} + \frac{C\frac{\partial^2}{(\partial\theta)^2}u_4}{24\,r|Cr-1|u_4}$$
$$- \frac{\frac{\partial}{\partial r}u_4}{12\,r|Cr-1|u_4} - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{24\,r^2\sin(\theta)u_4} - \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{24\,r^2u_4} - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{24\,r^2|Cr-1|\sin(\theta)u_4} - \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{24\,r^2|Cr-1|u_4} + \frac{C\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r|Cr-1|\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2|Cr-1|\sin(\theta)^2u_4}$$

$$\Psi_3 = -\frac{\sqrt{Cr+1}C\frac{\partial^2}{\partial r\partial\theta}u_4}{4\sqrt{|Cr-1|}u_4} + \frac{i\sqrt{Cr+1}C\frac{\partial^2}{\partial r\partial\phi}u_4}{4\sqrt{|Cr-1|}\sin(\theta)u_4} + \frac{\sqrt{Cr+1}C\frac{\partial}{\partial\theta}u_4}{4\,r\sqrt{|Cr-1|}u_4} + \frac{\sqrt{Cr+1}\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,r\sqrt{|Cr-1|}u_4}$$
$$- \frac{i\sqrt{Cr+1}C\frac{\partial}{\partial\phi}u_4}{4\,r\sqrt{|Cr-1|}\sin(\theta)u_4} - \frac{i\sqrt{Cr+1}\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,r\sqrt{|Cr-1|}\sin(\theta)u_4} - \frac{\sqrt{Cr+1}\frac{\partial}{\partial\theta}u_4}{4\,r^2\sqrt{|Cr-1|}u_4} + \frac{i\sqrt{Cr+1}\frac{\partial}{\partial\phi}u_4}{4\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}$$

$$\Psi_4 = \frac{C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r|Cr-1|\sin(\theta)u_4} - \frac{C\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r|Cr-1|u_4} - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2u_4} + \frac{i\,C\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)u_4} - \frac{i\,C\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)^2u_4} - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2|Cr-1|u_4}$$
$$- \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r^2\sin(\theta)u_4} + \frac{i\,\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r^2\sin(\theta)^2u_4} + \frac{C\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r|Cr-1|\sin(\theta)^2u_4} - \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r^2|Cr-1|\sin(\theta)u_4} + \frac{i\,\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r^2|Cr-1|\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2|Cr-1|\sin(\theta)^2u_4}$$

Newman-Penrose Equations

$$
\begin{aligned}
NP1 = & -\frac{C^3 r^3 \frac{\partial^2}{(\partial r)^2} u_4}{8\,|Cr-1| u_4} - \frac{3\,C^3 r^2 \frac{\partial}{\partial r} u_4}{8\,|Cr-1| u_4} - \frac{C^2 r^2 \frac{\partial^2}{(\partial r)^2} u_4}{8\,u_4} + \frac{5\,C^3 r}{8\,|Cr-1|} - \frac{3\,C^2 r \frac{\partial}{\partial r} u_4}{8\,u_4} + \frac{C^2 r^2 \frac{\partial^2}{(\partial r)^2} u_4}{8\,|Cr-1| u_4} + \frac{5}{8}\,C^2 + \frac{3\,C^2 r \frac{\partial}{\partial r} u_4}{8\,|Cr-1| u_4} - \frac{5\,C^2}{8\,|Cr-1|} \\
& + \frac{Cr \frac{\partial^2}{(\partial r)^2} u_4}{8\,|Cr-1| u_4} + \frac{C \frac{\partial}{\partial r} u_4}{4\,|Cr-1| u_4} + \frac{\frac{\partial^2}{(\partial r)^2} u_4}{8\,u_4} + \frac{\frac{\partial}{\partial r} u_4}{4\,r u_4} - \frac{\frac{\partial^2}{(\partial r)^2} u_4}{8\,|Cr-1| u_4} - \frac{C \cos(\theta) \frac{\partial}{\partial \theta} u_4}{8\,r|Cr-1| \sin(\theta) u_4} - \frac{C \frac{\partial^2}{(\partial \theta)^2} u_4}{8\,r|Cr-1| u_4} - \frac{\frac{\partial}{\partial r} u_4}{4\,r|Cr-1| u_4} \\
& - \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{8\,r^2 \sin(\theta) u_4} - \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{8\,r^2 u_4} + \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{8\,r^2|Cr-1| \sin(\theta) u_4} + \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{8\,r^2|Cr-1| u_4} - \frac{C \frac{\partial^2}{(\partial \phi)^2} u_4}{8\,r|Cr-1| \sin(\theta)^2 u_4} - \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{8\,r^2 \sin(\theta)^2 u_4} + \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{8\,r^2|Cr-1| \sin(\theta)^2 u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP2 = & \frac{C \cos(\theta) \frac{\partial}{\partial \theta} u_4}{8\,r|Cr-1| \sin(\theta) u_4} - \frac{C \frac{\partial^2}{(\partial \theta)^2} u_4}{8\,r|Cr-1| u_4} + \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{8\,r^2 \sin(\theta) u_4} - \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{8\,r^2 u_4} - \frac{i\,C \frac{\partial^2}{\partial \theta \partial \phi} u_4}{4\,r|Cr-1| \sin(\theta) u_4} + \frac{i\,C \cos(\theta) \frac{\partial}{\partial \phi} u_4}{4\,r|Cr-1| \sin(\theta)^2 u_4} - \frac{\cos(\theta) \frac{\partial}{\partial \theta} u_4}{8\,r^2|Cr-1| \sin(\theta) u_4} + \frac{\frac{\partial^2}{(\partial \theta)^2} u_4}{8\,r^2|Cr-1| u_4} \\
& - \frac{i\,\frac{\partial^2}{\partial \theta \partial \phi} u_4}{4\,r^2 \sin(\theta) u_4} + \frac{i\,\cos(\theta) \frac{\partial}{\partial \phi} u_4}{4\,r^2 \sin(\theta)^2 u_4} + \frac{C \frac{\partial^2}{(\partial \phi)^2} u_4}{8\,r|Cr-1| \sin(\theta)^2 u_4} + \frac{i\,\frac{\partial^2}{\partial \theta \partial \phi} u_4}{4\,r^2|Cr-1| \sin(\theta) u_4} - \frac{i\,\cos(\theta) \frac{\partial}{\partial \phi} u_4}{4\,r^2|Cr-1| \sin(\theta)^2 u_4} + \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{8\,r^2 \sin(\theta)^2 u_4} - \frac{\frac{\partial^2}{(\partial \phi)^2} u_4}{8\,r^2|Cr-1| \sin(\theta)^2 u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP3 = & \frac{\sqrt{Cr+1} C^3 r \frac{\partial}{\partial \theta} u_4}{4\,(C^2 r^2 - 1)\sqrt{|Cr-1|} u_4} + \frac{\sqrt{Cr+1} C^2 \sqrt{|Cr-1|} \frac{\partial}{\partial \theta} u_4}{4\,(C^2 r^2 - 1) u_4} + \frac{i\,\sqrt{Cr+1} C^3 r \frac{\partial}{\partial \phi} u_4}{4\,(C^2 r^2 - 1)\sqrt{|Cr-1|} \sin(\theta) u_4} \\
& - \frac{\sqrt{Cr+1} C^2 \frac{\partial}{\partial \theta} u_4}{4\,(C^2 r^2 - 1)\sqrt{|Cr-1|} u_4} + \frac{i\,\sqrt{Cr+1} C^2 \sqrt{|Cr-1|} \frac{\partial}{\partial \phi} u_4}{4\,(C^2 r^2 - 1) \sin(\theta) u_4} - \frac{i\,\sqrt{Cr+1} C^2 \frac{\partial}{\partial \phi} u_4}{4\,(C^2 r^2 - 1)\sqrt{|Cr-1|} \sin(\theta) u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP4 = & \frac{C^3 r \frac{\partial}{\partial \theta} u_4}{8\,\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|} u_4} + \frac{C^2 \sqrt{|Cr-1|} \frac{\partial}{\partial \theta} u_4}{8\,\sqrt{Cr+1}(Cr-1) u_4} - \frac{i\,C^3 r \frac{\partial}{\partial \phi} u_4}{8\,\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|} \sin(\theta) u_4} \\
& - \frac{C^2 \frac{\partial}{\partial \theta} u_4}{8\,\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|} u_4} - \frac{i\,C^2 \sqrt{|Cr-1|} \frac{\partial}{\partial \phi} u_4}{8\,\sqrt{Cr+1}(Cr-1) \sin(\theta) u_4} + \frac{i\,C^2 \frac{\partial}{\partial \phi} u_4}{8\,\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|} \sin(\theta) u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP5 = & \frac{C^3 r \frac{\partial}{\partial \theta} u_4}{8\,\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|} u_4} + \frac{C^2 \sqrt{|Cr-1|} \frac{\partial}{\partial \theta} u_4}{8\,\sqrt{Cr+1}(Cr-1) u_4} + \frac{i\,C^3 r \frac{\partial}{\partial \phi} u_4}{8\,\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|} \sin(\theta) u_4} \\
& - \frac{C^2 \frac{\partial}{\partial \theta} u_4}{8\,\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|} u_4} + \frac{i\,C^2 \sqrt{|Cr-1|} \frac{\partial}{\partial \phi} u_4}{8\,\sqrt{Cr+1}(Cr-1) \sin(\theta) u_4} - \frac{i\,C^2 \frac{\partial}{\partial \phi} u_4}{8\,\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|} \sin(\theta) u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP6 = & -\frac{17\,C^5r^7\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} - \frac{C^5r^6\frac{\partial}{\partial r}u_4}{16\,(C^2r^4-r^2)|Cr-1|u_4} - \frac{17\,C^4r^6\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^2r^4-r^2)u_4} - \frac{3\,C^5r^5}{16\,(C^2r^4-r^2)|Cr-1|} - \frac{C^4r^5\frac{\partial}{\partial r}u_4}{16\,(C^2r^4-r^2)u_4} + \frac{17\,C^4r^6\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} \\
& - \frac{3\,C^4r^4}{16\,(C^2r^4-r^2)} + \frac{C^4r^5\frac{\partial}{\partial r}u_4}{16\,(C^2r^4-r^2)|Cr-1|u_4} + \frac{3\,C^4r^4}{16\,(C^2r^4-r^2)|Cr-1|} + \frac{17\,C^3r^5\frac{\partial^2}{(\partial r)^2}u_4}{24\,(C^2r^4-r^2)|Cr-1|u_4} + \frac{13\,C^3r^4\frac{\partial}{\partial r}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} + \frac{17\,C^2r^4\frac{\partial^2}{(\partial r)^2}u_4}{24\,(C^2r^4-r^2)u_4} \\
& - \frac{5\,C^3r^3}{16\,(C^2r^4-r^2)|Cr-1|} + \frac{13\,C^2r^3\frac{\partial}{\partial r}u_4}{48\,(C^2r^4-r^2)u_4} - \frac{17\,C^2r^4\frac{\partial^2}{(\partial r)^2}u_4}{24\,(C^2r^4-r^2)|Cr-1|u_4} + \frac{5\,C^3r^3\cos(\theta)\frac{\partial}{\partial\theta}u_4}{48\,(C^2r^4-r^2)|Cr-1|\sin(\theta)u_4} + \frac{5\,C^3r^3\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} \\
& - \frac{5\,C^2r^2}{16\,(C^2r^4-r^2)} - \frac{13\,C^2r^3\frac{\partial}{\partial r}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} + \frac{5\,C^2r^2\cos(\theta)\frac{\partial}{\partial\theta}u_4}{48\,(C^2r^4-r^2)\sin(\theta)u_4} + \frac{5\,C^2r^2\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,(C^2r^4-r^2)u_4} + \frac{5\,C^2r^2}{16\,(C^2r^4-r^2)|Cr-1|} - \frac{17\,Cr^3\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} \\
& - \frac{5\,C^2r^2\cos(\theta)\frac{\partial}{\partial\theta}u_4}{48\,(C^2r^4-r^2)|Cr-1|\sin(\theta)u_4} - \frac{5\,C^2r^2\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} + \frac{5\,C^3r^3\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|\sin(\theta)^2u_4} - \frac{5\,Cr^2\frac{\partial}{\partial r}u_4}{24\,(C^2r^4-r^2)|Cr-1|u_4} - \frac{17\,r^2\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^2r^4-r^2)u_4} \\
& + \frac{5\,C^2r^2\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,(C^2r^4-r^2)\sin(\theta)^2u_4} - \frac{5\,r\frac{\partial}{\partial r}u_4}{24\,(C^2r^4-r^2)u_4} + \frac{17\,r^2\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} - \frac{5\,Cr\cos(\theta)\frac{\partial}{\partial\theta}u_4}{48\,(C^2r^4-r^2)|Cr-1|\sin(\theta)u_4} - \frac{5\,Cr\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} \\
& - \frac{5\,C^2r^2\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|\sin(\theta)^2u_4} + \frac{5\,r\frac{\partial}{\partial r}u_4}{24\,(C^2r^4-r^2)|Cr-1|u_4} - \frac{5\,\cos(\theta)\frac{\partial}{\partial\theta}u_4}{48\,(C^2r^4-r^2)\sin(\theta)u_4} - \frac{5\,\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,(C^2r^4-r^2)u_4} + \frac{5\,\cos(\theta)\frac{\partial}{\partial\theta}u_4}{48\,(C^2r^4-r^2)|Cr-1|\sin(\theta)u_4} \\
& + \frac{5\,\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|u_4} - \frac{5\,Cr\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|\sin(\theta)^2u_4} - \frac{5\,\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,(C^2r^4-r^2)\sin(\theta)^2u_4} + \frac{5\,\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,(C^2r^4-r^2)|Cr-1|\sin(\theta)^2u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP8 = & \frac{C^3r^3\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} + \frac{C^3r^2\frac{\partial}{\partial r}u_4}{2\,|Cr-1|u_4} + \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{12\,u_4} + \frac{C^3r}{4\,|Cr-1|} + \frac{C^2r\frac{\partial}{\partial r}u_4}{2\,u_4} - \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} + \frac{1}{4}\,C^2 - \frac{C^2r\frac{\partial}{\partial r}u_4}{2\,|Cr-1|u_4} - \frac{C^2}{4\,|Cr-1|} \\
& - \frac{Cr\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} - \frac{5\,C\frac{\partial}{\partial r}u_4}{12\,|Cr-1|u_4} - \frac{\frac{\partial^2}{(\partial r)^2}u_4}{12\,u_4} - \frac{5\,\frac{\partial}{\partial r}u_4}{12\,ru_4} + \frac{\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} + \frac{C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{24\,r|Cr-1|\sin(\theta)u_4} + \frac{C\frac{\partial^2}{(\partial\theta)^2}u_4}{24\,r|Cr-1|u_4} + \frac{5\,\frac{\partial}{\partial r}u_4}{12\,r|Cr-1|u_4} \\
& + \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{24\,r^2\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{24\,r^2u_4} - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{24\,r^2|Cr-1|\sin(\theta)u_4} - \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{24\,r^2|Cr-1|u_4} + \frac{C\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r|Cr-1|\sin(\theta)^2u_4} + \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2|Cr-1|\sin(\theta)^2u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP9 = & -\frac{\sqrt{Cr+1}C^3r\frac{\partial}{\partial\theta}u_4}{4\,(C^2r^2-1)\sqrt{|Cr-1|}u_4} - \frac{\sqrt{Cr+1}C^2\sqrt{|Cr-1|}\frac{\partial}{\partial\theta}u_4}{4\,(C^2r^2-1)u_4} + \frac{i\,\sqrt{Cr+1}C^3r\frac{\partial}{\partial\phi}u_4}{4\,(C^2r^2-1)\sqrt{|Cr-1|}\sin(\theta)u_4} \\
& + \frac{\sqrt{Cr+1}C^2\frac{\partial}{\partial\theta}u_4}{4\,(C^2r^2-1)\sqrt{|Cr-1|}u_4} + \frac{i\,\sqrt{Cr+1}C^2\sqrt{|Cr-1|}\frac{\partial}{\partial\phi}u_4}{4\,(C^2r^2-1)\sin(\theta)u_4} - \frac{i\,\sqrt{Cr+1}C^2\frac{\partial}{\partial\phi}u_4}{4\,(C^2r^2-1)\sqrt{|Cr-1|}\sin(\theta)u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP10 = & -\frac{C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r|Cr-1|\sin(\theta)u_4} + \frac{C\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r|Cr-1|u_4} - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2u_4} - \frac{i\,C\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)u_4} + \frac{i\,C\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)^2u_4} + \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4} - \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2|Cr-1|u_4} \\
& - \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r^2\sin(\theta)u_4} + \frac{i\,\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r^2\sin(\theta)^2u_4} - \frac{C\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r|Cr-1|\sin(\theta)^2u_4} + \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r^2|Cr-1|\sin(\theta)u_4} - \frac{i\,\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r^2|Cr-1|\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2\sin(\theta)^2u_4} + \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2|Cr-1|\sin(\theta)^2u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP12 = & -\frac{C^3r^3\frac{\partial^2}{(\partial r)^2}u_4}{48\,|Cr-1|u_4} - \frac{C^3r^2\frac{\partial}{\partial r}u_4}{16\,|Cr-1|u_4} - \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{48\,u_4} + \frac{9\,C^3r}{16\,|Cr-1|} - \frac{C^2r\frac{\partial}{\partial r}u_4}{16\,u_4} + \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{48\,|Cr-1|u_4} + \frac{9}{16}\,C^2 + \frac{C^2r\frac{\partial}{\partial r}u_4}{16\,|Cr-1|u_4} - \frac{9\,C^2}{16\,|Cr-1|} + \frac{Cr\frac{\partial^2}{(\partial r)^2}u_4}{48\,|Cr-1|u_4} \\
& + \frac{C\frac{\partial}{\partial r}u_4}{24\,|Cr-1|u_4} + \frac{\frac{\partial^2}{(\partial r)^2}u_4}{48\,u_4} - \frac{C}{2\,r|Cr-1|} + \frac{\frac{\partial}{\partial r}u_4}{24\,ru_4} - \frac{\frac{\partial^2}{(\partial r)^2}u_4}{48\,|Cr-1|u_4} + \frac{C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{48\,r|Cr-1|\sin(\theta)u_4} + \frac{C\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r|Cr-1|u_4} - \frac{1}{2\,r^2} - \frac{\frac{\partial}{\partial r}u_4}{24\,r|Cr-1|u_4} + \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{48\,r^2\sin(\theta)u_4} \\
& + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^2u_4} + \frac{1}{2\,r^2|Cr-1|} - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{48\,r^2|Cr-1|\sin(\theta)u_4} - \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^2|Cr-1|u_4} + \frac{C\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,r|Cr-1|\sin(\theta)^2u_4} + \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,r^2\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,r^2|Cr-1|\sin(\theta)^2u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP14 = & -\frac{C^3r^3\frac{\partial^2}{(\partial r)^2}u_4}{8\,|Cr-1|u_4} - \frac{3\,C^3r^2\frac{\partial}{\partial r}u_4}{8\,|Cr-1|u_4} - \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{8\,u_4} + \frac{5\,C^3r}{8\,|Cr-1|} - \frac{3\,C^2r\frac{\partial}{\partial r}u_4}{8\,u_4} + \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{8\,|Cr-1|u_4} + \frac{5}{8}\,C^2 + \frac{3\,C^2r\frac{\partial}{\partial r}u_4}{8\,|Cr-1|u_4} - \frac{5\,C^2}{8\,|Cr-1|} \\
& + \frac{Cr\frac{\partial^2}{(\partial r)^2}u_4}{8\,|Cr-1|u_4} + \frac{C\frac{\partial}{\partial r}u_4}{4\,|Cr-1|u_4} + \frac{\frac{\partial^2}{(\partial r)^2}u_4}{8\,u_4} + \frac{\frac{\partial}{\partial r}u_4}{4\,ru_4} - \frac{\frac{\partial^2}{(\partial r)^2}u_4}{8\,|Cr-1|u_4} - \frac{C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r|Cr-1|\sin(\theta)u_4} - \frac{C\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r|Cr-1|u_4} - \frac{\frac{\partial}{\partial r}u_4}{4\,r|Cr-1|u_4} \\
& - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2\sin(\theta)u_4} - \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2u_4} + \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2|Cr-1|u_4} - \frac{C\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r|Cr-1|\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2\sin(\theta)^2u_4} + \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2|Cr-1|\sin(\theta)^2u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP15 = & -\frac{C^3r\frac{\partial}{\partial\theta}u_4}{8\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|}u_4} - \frac{C^2\sqrt{|Cr-1|}\frac{\partial}{\partial\theta}u_4}{8\sqrt{Cr+1}(Cr-1)u_4} - \frac{i\,C^3r\frac{\partial}{\partial\phi}u_4}{8\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|}\sin(\theta)u_4} \\
& + \frac{C^2\frac{\partial}{\partial\theta}u_4}{8\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|}u_4} - \frac{i\,C^2\sqrt{|Cr-1|}\frac{\partial}{\partial\phi}u_4}{8\sqrt{Cr+1}(Cr-1)\sin(\theta)u_4} + \frac{i\,C^2\frac{\partial}{\partial\phi}u_4}{8\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|}\sin(\theta)u_4}
\end{aligned}
$$

$$
\begin{aligned}
NP17 = & -\frac{C^3r^3\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} - \frac{C^3r^2\frac{\partial}{\partial r}u_4}{2\,|Cr-1|u_4} - \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{12\,u_4} - \frac{C^3r}{4\,|Cr-1|} - \frac{C^2r\frac{\partial}{\partial r}u_4}{2\,u_4} + \frac{C^2r^2\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} - \frac{1}{4}\,C^2 + \frac{C^2r\frac{\partial}{\partial r}u_4}{2\,|Cr-1|u_4} + \frac{C^2}{4\,|Cr-1|} \\
& + \frac{Cr\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} + \frac{5\,C\frac{\partial}{\partial r}u_4}{12\,|Cr-1|u_4} + \frac{\frac{\partial^2}{(\partial r)^2}u_4}{12\,u_4} + \frac{5\,\frac{\partial}{\partial r}u_4}{12\,ru_4} - \frac{\frac{\partial^2}{(\partial r)^2}u_4}{12\,|Cr-1|u_4} - \frac{C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{24\,r|Cr-1|\sin(\theta)u_4} - \frac{C\frac{\partial^2}{(\partial\theta)^2}u_4}{24\,r|Cr-1|u_4} - \frac{5\,\frac{\partial}{\partial r}u_4}{12\,r|Cr-1|u_4} \\
& - \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{24\,r^2\sin(\theta)u_4} - \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{24\,r^2u_4} + \frac{\cos(\theta)\frac{\partial}{\partial\theta}u_4}{24\,r^2|Cr-1|\sin(\theta)u_4} + \frac{\frac{\partial^2}{(\partial\theta)^2}u_4}{24\,r^2|Cr-1|u_4} - \frac{C\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r|Cr-1|\sin(\theta)^2u_4} - \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2\sin(\theta)^2u_4} + \frac{\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2|Cr-1|\sin(\theta)^2u_4}
\end{aligned}
$$

$$NP18 = \frac{C^3 r \frac{\partial}{\partial\theta} u_4}{8\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|}u_4} + \frac{C^2\sqrt{|Cr-1|}\frac{\partial}{\partial\theta}u_4}{8\sqrt{Cr+1}(Cr-1)u_4} - \frac{i\,C^3 r\frac{\partial}{\partial\phi}u_4}{8\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|}\sin(\theta)u_4}$$
$$- \frac{C^2\frac{\partial}{\partial\theta}u_4}{8\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|}u_4} - \frac{i\,C^2\sqrt{|Cr-1|}\frac{\partial}{\partial\phi}u_4}{8\sqrt{Cr+1}(Cr-1)\sin(\theta)u_4} + \frac{i\,C^2\frac{\partial}{\partial\phi}u_4}{8\sqrt{Cr+1}(Cr-1)\sqrt{|Cr-1|}\sin(\theta)u_4}$$

Bianchi Identities

$$BI1 = -\frac{\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,|Cr-1|u_4} - \frac{\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,u_4} - \frac{i\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} - \frac{3\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,|Cr-1|u_4} - \frac{\sqrt{2}C^3r\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,|Cr-1|u_4^2} + \frac{\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,|Cr-1|u_4} - \frac{i\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,\sin(\theta)u_4} - \frac{3\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,u_4}$$
$$- \frac{3i\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}C^3\frac{\partial}{\partial\theta}u_4}{4\,|Cr-1|u_4} - \frac{\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,u_4^2} - \frac{i\sqrt{2}C^3r\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4^2} + \frac{i\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} + \frac{3\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,|Cr-1|u_4} - \frac{3i\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{16\,\sin(\theta)u_4} + \frac{\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{4\,ru_4}$$
$$+ \frac{\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,|Cr-1|u_4^2} + \frac{i\sqrt{2}C^3\frac{\partial}{\partial\phi}u_4}{4\,|Cr-1|\sin(\theta)u_4} - \frac{i\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,\sin(\theta)u_4^2} + \frac{\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,|Cr-1|u_4} + \frac{3i\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{4\,r|Cr-1|u_4} + \frac{i\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{4\,r\sin(\theta)u_4} + \frac{i\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4^2}$$
$$+ \frac{\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,ru_4} + \frac{i\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}C\frac{\partial^2}{\partial r\partial\theta}u_4}{8\,r|Cr-1|u_4} + \frac{\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r|Cr-1|u_4^2} - \frac{i\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,r|Cr-1|u_4} + \frac{i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,r\sin(\theta)u_4} + \frac{\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{8\,r^2u_4} + \frac{i\sqrt{2}C\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)u_4}$$
$$- \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4}{8\,r^2|Cr-1|u_4} + \frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2u_4^2} + \frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{16\,r^2|Cr-1|\sin(\theta)u_4^2} + \frac{i\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,r|Cr-1|\sin(\theta)u_4^2} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,r|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{8\,r^2|Cr-1|u_4} + \frac{i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r^2\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{8\,r^3u_4}$$
$$- \frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2|Cr-1|u_4^2} + \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{16\,r^3\sin(\theta)u_4^2} - \frac{i\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2} - \frac{i\sqrt{2}C\frac{\partial}{\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4} + \frac{i\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2\sin(\theta)u_4^2} + \frac{i\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2} + \frac{i\sqrt{2}C\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2}$$
$$- \frac{i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{8\,r^3|Cr-1|u_4} - \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{16\,r^3|Cr-1|\sin(\theta)u_4^2} - \frac{i\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^3\sin(\theta)u_4^2} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{8\,r^3\sin(\theta)u_4} - \frac{i\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2} + \frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3\sin(\theta)^2u_4^2} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3\sin(\theta)u_4^2}$$
$$- \frac{\sqrt{2}C\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2} + \frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^2|Cr-1|\sin(\theta)^3u_4^2} + \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2} + \frac{i\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)u_4^2} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{8\,r^3|Cr-1|\sin(\theta)u_4} - \frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)^2u_4^2}$$
$$- \frac{i\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)u_4^2} - \frac{\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3\sin(\theta)^2u_4^2} + \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^3\sin(\theta)^3u_4^2} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3\sin(\theta)^2u_4^2} + \frac{\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)^2u_4^2} - \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^3|Cr-1|\sin(\theta)^3u_4^2} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3|Cr-1|\sin(\theta)^2u_4^2}$$

$$
\begin{aligned}
BI2 = &-\frac{\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{16\,|Cr-1|^{\frac{3}{2}}u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,|Cr-1|^{\frac{3}{2}}u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,r\sqrt{|Cr-1|}\sin(\theta)u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{16\,r\sqrt{|Cr-1|}u_4}\\
&+\frac{i\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{8\,|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{i\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r\sqrt{|Cr-1|}\sin(\theta)u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r|Cr-1|^{\frac{3}{2}}u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r\sqrt{|Cr-1|}u_4^2}\\
&+\frac{i\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}-\frac{i\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{8\,|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{8\,r|Cr-1|^{\frac{3}{2}}u_4}+\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{8\,r\sqrt{|Cr-1|}\sin(\theta)u_4}-\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r\sqrt{|Cr-1|}\sin(\theta)^2u_4}\\
&-\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{16\,|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2\sqrt{|Cr-1|}u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r|Cr-1|^{\frac{3}{2}}u_4^2}-\frac{i\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r\sqrt{|Cr-1|}\sin(\theta)u_4^2}\\
&+\frac{i\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial}{\partial\phi}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{8\,r\sqrt{|Cr-1|}\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}+\frac{\sqrt{2}C\cos(\theta)\frac{\partial^2}{\partial t\partial\theta}u_4}{16\,r|Cr-1|\sin(\theta)u_4^2}-\frac{\sqrt{2}C\frac{\partial^3}{\partial t(\partial\theta)^2}u_4}{16\,r|Cr-1|u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{16\,r^2\sqrt{|Cr-1|}u_4}\\
&-\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{4\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{16\,r\sqrt{|Cr-1|}\sin(\theta)^2u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\theta}u_4}{16\,r|Cr-1|\sin(\theta)u_4^3}+\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}u_4}\\
&+\frac{\sqrt{2}C\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r|Cr-1|u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2\sqrt{|Cr-1|}u_4^2}-\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial\theta\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}-\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}+\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4}+\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{4\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}\\
&-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r\sqrt{|Cr-1|}\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial t\partial\theta}u_4}{16\,r^2\sin(\theta)u_4^2}-\frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\theta)^2}u_4}{16\,r^2u_4^2}-\frac{i\sqrt{2}C\frac{\partial^3}{\partial t\partial\theta\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)u_4^2}+\frac{i\sqrt{2}C\cos(\theta)\frac{\partial^2}{\partial t\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}u_4}-\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}\\
&+\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\,r^3\sqrt{|Cr-1|}\sin(\theta)u_4}-\frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2\sin(\theta)u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^3\sqrt{|Cr-1|}u_4}+\frac{\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2u_4^3}\\
&+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}u_4^2}+\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{i\sqrt{2}C\frac{\partial}{\partial t}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)u_4^3}-\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4^2}-\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{i\sqrt{2}C\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)^2u_4^3}+\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4^2}\\
&+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4}+\frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial t\partial\theta}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2}+\frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\theta)^2}u_4}{16\,r^2|Cr-1|u_4^2}-\frac{i\sqrt{2}\frac{\partial^3}{\partial t\partial\theta\partial\phi}u_4}{8\,r^2\sin(\theta)u_4^2}+\frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial t\partial\phi}u_4}{8\,r^2\sin(\theta)^2u_4^2}+\frac{\sqrt{2}C\frac{\partial^3}{\partial t(\partial\phi)^2}u_4}{16\,r|Cr-1|\sin(\theta)^2u_4^2}+\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}\\
&-\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\,r^3|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^3|Cr-1|^{\frac{3}{2}}u_4}-\frac{\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2|Cr-1|u_4^3}+\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^3\sqrt{|Cr-1|}\sin(\theta)u_4}\\
&+\frac{i\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^2\sin(\theta)u_4^3}+\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}-\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial\phi}u_4}{8\,r^3\sqrt{|Cr-1|}\sin(\theta)^2u_4}-\frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\phi}u_4}{8\,r^2\sin(\theta)^2u_4^3}-\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{\sqrt{2}C\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r|Cr-1|\sin(\theta)^2u_4^3}\\
&+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4^2}+\frac{i\sqrt{2}\frac{\partial^3}{\partial t\partial\theta\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4^2}-\frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial t\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}u_4}{16\,r^2\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{i\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^3|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{i\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4^3}+\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial\phi}u_4}{8\,r^3|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}\\
&+\frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)^2u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3\sqrt{|Cr-1|}\sin(\theta)^2u_4}-\frac{\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2\sin(\theta)^2u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^3}
\end{aligned}
$$

$$
\begin{aligned}
BI3 = &-\frac{\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{16\,|Cr-1|^{\frac{3}{2}}u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,|Cr-1|^{\frac{3}{2}}u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,r\sqrt{|Cr-1|}\sin(\theta)u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{16\,r\sqrt{|Cr-1|}u_4}\\
&-\frac{i\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{8\,|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{i\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r\sqrt{|Cr-1|}\sin(\theta)u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r|Cr-1|^{\frac{3}{2}}u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r\sqrt{|Cr-1|}u_4^2}\\
&-\frac{i\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}+\frac{i\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{8\,|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{8\,r|Cr-1|^{\frac{3}{2}}u_4}-\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{8\,r\sqrt{|Cr-1|}\sin(\theta)u_4}+\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r\sqrt{|Cr-1|}\sin(\theta)^2u_4}\\
&-\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{16\,|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2\sqrt{|Cr-1|}u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r|Cr-1|^{\frac{3}{2}}u_4^2}+\frac{i\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r\sqrt{|Cr-1|}\sin(\theta)u_4^2}\\
&-\frac{i\sqrt{2}\sqrt{Cr+1}C^2\cos(\theta)\frac{\partial}{\partial\phi}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{8\,r\sqrt{|Cr-1|}\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}-\frac{\sqrt{2}C\cos(\theta)\frac{\partial^2}{\partial t\partial\theta}u_4}{16\,r|Cr-1|\sin(\theta)u_4^2}+\frac{\sqrt{2}C\frac{\partial^3}{\partial t(\partial\theta)^2}u_4}{16\,r|Cr-1|u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{16\,r^2\sqrt{|Cr-1|}u_4}\\
&+\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{4\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{16\,r\sqrt{|Cr-1|}\sin(\theta)^2u_4}-\frac{\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial\theta}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\theta}u_4}{16\,r|Cr-1|\sin(\theta)u_4^3}+\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{(\partial\theta)^2}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}u_4}\\
&-\frac{\sqrt{2}C\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r|Cr-1|u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2\sqrt{|Cr-1|}u_4^2}+\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial\theta\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}+\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}-\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4}-\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{4\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}C^2\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}\\
&-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r\sqrt{|Cr-1|}\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial t\partial\theta}u_4}{16\,r^2\sin(\theta)u_4^2}+\frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\theta)^2}u_4}{16\,r^2u_4^2}-\frac{i\sqrt{2}C\frac{\partial^3}{\partial t\partial\theta\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)u_4^2}+\frac{i\sqrt{2}C\cos(\theta)\frac{\partial^2}{\partial t\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}u_4}+\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4}\\
&-\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{8\,r|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\,r^3\sqrt{|Cr-1|}\sin(\theta)u_4}+\frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2\sin(\theta)u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^3\sqrt{|Cr-1|}u_4}-\frac{\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2u_4^3}\\
&+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}u_4^2}-\frac{i\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{\partial\theta\partial\phi}u_4}{4\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}+\frac{i\sqrt{2}C\frac{\partial}{\partial t}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)u_4^3}+\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)u_4^2}+\frac{i\sqrt{2}\sqrt{Cr+1}C\cos(\theta)\frac{\partial}{\partial\phi}u_4}{4\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{i\sqrt{2}C\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)^2u_4^3}-\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{8\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4^2}\\
&+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial t\partial\theta}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2}-\frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\theta)^2}u_4}{16\,r^2|Cr-1|u_4^2}-\frac{i\sqrt{2}\frac{\partial^3}{\partial t\partial\theta\partial\phi}u_4}{8\,r^2\sin(\theta)u_4^2}+\frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial t\partial\phi}u_4}{8\,r^2\sin(\theta)^2u_4^2}-\frac{\sqrt{2}C\frac{\partial^3}{\partial t(\partial\phi)^2}u_4}{16\,r|Cr-1|\sin(\theta)^2u_4^2}-\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r\partial\theta\partial\phi}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}\\
&+\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4}+\frac{\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\,r^3|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^3|Cr-1|^{\frac{3}{2}}u_4}+\frac{\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{16\,r^2|Cr-1|u_4^3}-\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^3\sqrt{|Cr-1|}\sin(\theta)u_4}\\
&+\frac{i\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^2\sin(\theta)u_4^3}-\frac{i\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4^2}+\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial\phi}u_4}{8\,r^3\sqrt{|Cr-1|}\sin(\theta)^2u_4}-\frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\phi}u_4}{8\,r^2\sin(\theta)^2u_4^3}+\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}C\frac{\partial^2}{(\partial\phi)^2}u_4}{8\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{\sqrt{2}C\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r|Cr-1|\sin(\theta)^2u_4^3}\\
&+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2\sqrt{|Cr-1|}\sin(\theta)^2u_4^2}+\frac{i\sqrt{2}\frac{\partial^3}{\partial t\partial\theta\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4^2}-\frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial t\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}u_4}{16\,r^2\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}+\frac{\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^3|Cr-1|^{\frac{3}{2}}\sin(\theta)u_4}-\frac{i\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4^3}-\frac{i\sqrt{2}\sqrt{Cr+1}\cos(\theta)\frac{\partial}{\partial\phi}u_4}{8\,r^3|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}\\
&+\frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial t}u_4\frac{\partial}{\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)^2u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3\sqrt{|Cr-1|}\sin(\theta)^2u_4}+\frac{\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2\sin(\theta)^2u_4^3}-\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2}+\frac{\sqrt{2}\sqrt{Cr+1}\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3|Cr-1|^{\frac{3}{2}}\sin(\theta)^2u_4}-\frac{\sqrt{2}\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^3}
\end{aligned}
$$

$$
\begin{aligned}
BI4 = {} & \frac{\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,|Cr-1|u_4} + \frac{\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,u_4} - \frac{i\,\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} + \frac{3\,\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,|Cr-1|u_4} + \frac{\sqrt{2}C^3r\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,|Cr-1|u_4^2} - \frac{\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,|Cr-1|u_4} - \frac{i\,\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,\sin(\theta)u_4} + \frac{3\,\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,u_4} \\
& - \frac{3i\,\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}C^3\frac{\partial}{\partial\theta}u_4}{4\,|Cr-1|u_4} + \frac{\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,u_4^2} - \frac{i\,\sqrt{2}C^3r\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4^2} + \frac{i\,\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} - \frac{3\,\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,|Cr-1|u_4} - \frac{3i\,\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{16\,\sin(\theta)u_4} - \frac{\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{4\,ru_4} \\
& - \frac{\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,|Cr-1|u_4^2} + \frac{i\,\sqrt{2}C^3\frac{\partial}{\partial\phi}u_4}{4\,|Cr-1|\sin(\theta)u_4} - \frac{i\,\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,\sin(\theta)u_4^2} - \frac{\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,|Cr-1|u_4} + \frac{3i\,\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{4\,r|Cr-1|u_4} + \frac{i\,\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{4\,r\sin(\theta)u_4} + \frac{i\,\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4^2} \\
& - \frac{\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,ru_4} + \frac{i\,\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}C\frac{\partial^2}{\partial r\partial\theta}u_4}{8\,r|Cr-1|u_4} - \frac{\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r|Cr-1|u_4^2} - \frac{i\,\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{16\,r|Cr-1|u_4} + \frac{i\,\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,r\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{8\,r^2u_4} + \frac{i\,\sqrt{2}C\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)u_4} \\
& + \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4}{8\,r^2|Cr-1|u_4} - \frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2u_4^2} - \frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{16\,r^2|Cr-1|\sin(\theta)u_4^2} + \frac{i\,\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,r|Cr-1|\sin(\theta)u_4^2} - \frac{i\,\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{16\,r|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{8\,r^2|Cr-1|u_4} + \frac{i\,\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r^2\sin(\theta)u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{8\,r^3u_4} \\
& + \frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2|Cr-1|u_4^2} - \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{16\,r^3\sin(\theta)u_4^2} - \frac{i\,\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2} - \frac{i\,\sqrt{2}C\frac{\partial}{\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4} + \frac{i\,\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2\sin(\theta)u_4^2} + \frac{i\,\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2} + \frac{i\,\sqrt{2}C\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2} \\
& - \frac{i\,\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{8\,r^2|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{8\,r^3|Cr-1|u_4} + \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{16\,r^3|Cr-1|\sin(\theta)u_4^2} - \frac{i\,\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^3\sin(\theta)u_4^2} - \frac{i\,\sqrt{2}\frac{\partial}{\partial\phi}u_4}{8\,r^3\sin(\theta)u_4} - \frac{i\,\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2} + \frac{i\,\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3\sin(\theta)^2u_4^2} + \frac{i\,\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3\sin(\theta)u_4^2} \\
& + \frac{\sqrt{2}C\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2} - \frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^2|Cr-1|\sin(\theta)^3u_4^2} - \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2} + \frac{i\,\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)u_4^2} + \frac{i\,\sqrt{2}\frac{\partial}{\partial\phi}u_4}{8\,r^3|Cr-1|\sin(\theta)u_4} - \frac{i\,\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)^2u_4^2} \\
& - \frac{i\,\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)u_4^2} + \frac{\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3\sin(\theta)^2u_4^2} - \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^3\sin(\theta)^3u_4^2} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3\sin(\theta)^2u_4^2} - \frac{\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)^2u_4^2} + \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^3|Cr-1|\sin(\theta)^3u_4^2} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3|Cr-1|\sin(\theta)^2u_4^2}
\end{aligned}
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\begin{aligned}
BI5 = {} & \frac{\sqrt{2}\sqrt{Cr+1}C^5r^8\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} + \frac{5\,\sqrt{2}\sqrt{Cr+1}C^5r^8\sqrt{|Cr-1|}\frac{\partial^3}{(\partial r)^3}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4} - \frac{\sqrt{2}\sqrt{Cr+1}C^5r^7\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4^2}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} + \frac{13\,\sqrt{2}\sqrt{Cr+1}C^5r^7\sqrt{|Cr-1|}\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4} \\
& + \frac{\sqrt{2}\sqrt{Cr+1}C^4r^7|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} + \frac{5\,\sqrt{2}\sqrt{Cr+1}C^4r^7|Cr-1|^{\frac{3}{2}}\frac{\partial^3}{(\partial r)^3}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4} - \frac{\sqrt{2}C^5r^8\frac{\partial^3}{\partial t(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} - \frac{3\,\sqrt{2}\sqrt{Cr+1}C^5r^6\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4} \\
& - \frac{\sqrt{2}\sqrt{Cr+1}C^4r^6|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4^2}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} + \frac{13\,\sqrt{2}\sqrt{Cr+1}C^4r^6|Cr-1|^{\frac{3}{2}}\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4} + \frac{\sqrt{2}C^5r^8\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^3} - \frac{\sqrt{2}\sqrt{Cr+1}C^4r^7\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} \\
& - \frac{5\,\sqrt{2}\sqrt{Cr+1}C^4r^7\sqrt{|Cr-1|}\frac{\partial^3}{(\partial r)^3}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4} + \frac{\sqrt{2}\sqrt{Cr+1}C^5r^5\sqrt{|Cr-1|}}{4\,(C^3r^6-C^2r^5-Cr^4+r^3)} + \frac{\sqrt{2}C^5r^7\frac{\partial^2}{\partial t\partial r}u_4}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} - \frac{\sqrt{2}C^4r^7|Cr-1|\frac{\partial^3}{\partial t(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} - \frac{3\,\sqrt{2}\sqrt{Cr+1}C^4r^5|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4} \\
& - \frac{\sqrt{2}C^5r^7\frac{\partial}{\partial t}u_4\frac{\partial}{\partial r}u_4}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^3} + \frac{\sqrt{2}\sqrt{Cr+1}C^4r^6\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4^2}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} - \frac{13\,\sqrt{2}\sqrt{Cr+1}C^4r^6\sqrt{|Cr-1|}\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4} + \frac{\sqrt{2}C^4r^7|Cr-1|\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^3} \\
& + \frac{\sqrt{2}\sqrt{Cr+1}C^4r^4|Cr-1|^{\frac{3}{2}}}{4\,(C^3r^6-C^2r^5-Cr^4+r^3)} + \frac{\sqrt{2}C^4r^6|Cr-1|\frac{\partial^2}{\partial t\partial r}u_4}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} + \frac{\sqrt{2}C^4r^7\frac{\partial^3}{\partial t(\partial r)^2}u_4}{48\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^2} + \frac{3\,\sqrt{2}\sqrt{Cr+1}C^4r^5\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4} - \frac{\sqrt{2}C^4r^6|Cr-1|\frac{\partial}{\partial t}u_4\frac{\partial}{\partial r}u_4}{16\,(C^3r^6-C^2r^5-Cr^4+r^3)u_4^3} + \ldots
\end{aligned}
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$$
\begin{aligned}
BI6 = &-\frac{\sqrt{Cr+1}C^5r^8\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^2}-\frac{5\sqrt{Cr+1}C^5r^8\sqrt{|Cr-1|}\frac{\partial^3}{(\partial r)^3}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4}+\frac{\sqrt{Cr+1}C^5r^7\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4^2}{8\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^2}\\
&-\frac{13\sqrt{Cr+1}C^5r^7\sqrt{|Cr-1|}\frac{\partial^2}{(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4}-\frac{\sqrt{Cr+1}C^4r^7|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^2}-\frac{5\sqrt{Cr+1}C^4r^7|Cr-1|^{\frac{3}{2}}\frac{\partial^3}{(\partial r)^3}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4}\\
&-\frac{C^5r^8\frac{\partial^3}{\partial t(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^2}+\frac{3\sqrt{Cr+1}C^5r^6\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4}{8\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4}+\frac{\sqrt{Cr+1}C^4r^6|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4^2}{8\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^2}-\frac{13\sqrt{Cr+1}C^4r^6|Cr-1|^{\frac{3}{2}}\frac{\partial^2}{(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4}\\
&+\frac{C^5r^8\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^3}+\frac{\sqrt{Cr+1}C^4r^7\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^2}+\frac{5\sqrt{Cr+1}C^4r^7\sqrt{|Cr-1|}\frac{\partial^3}{(\partial r)^3}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4}-\frac{\sqrt{Cr+1}C^5r^5\sqrt{|Cr-1|}}{2\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)}\\
&+\frac{C^5r^7\frac{\partial^2}{\partial t\partial r}u_4}{8\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^2}-\frac{C^4r^7|Cr-1|\frac{\partial^3}{\partial t(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^2}+\frac{3\sqrt{Cr+1}C^4r^5|Cr-1|^{\frac{3}{2}}\frac{\partial}{\partial r}u_4}{8\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4}-\frac{C^5r^7\frac{\partial}{\partial t}u_4\frac{\partial}{\partial r}u_4}{8\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^3}\\
&-\frac{\sqrt{Cr+1}C^4r^6\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4^2}{8\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^2}+\frac{13\sqrt{Cr+1}C^4r^6\sqrt{|Cr-1|}\frac{\partial^2}{(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4}+\frac{C^4r^7|Cr-1|\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{24\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)u_4^3}-\frac{\sqrt{Cr+1}C^4r^4|Cr-1|^{\frac{3}{2}}}{2\,(\sqrt{2}C^3r^6-\sqrt{2}C^2r^5-\sqrt{2}Cr^4+\sqrt{2}r^3)}
\end{aligned}
$$

$$
\begin{aligned}
BI7 = &\frac{\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,|Cr-1|u_4}+\frac{\sqrt{2}C^3r^2\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,|Cr-1|u_4^2}+\frac{\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,u_4}-\frac{i\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,|Cr-1|\sin(\theta)u_4}+\frac{\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,|Cr-1|u_4}+\frac{\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,u_4^2}-\frac{i\sqrt{2}C^3r^2\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,|Cr-1|\sin(\theta)u_4^2}-\frac{\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,|Cr-1|u_4}-\frac{i\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,\sin(\theta)u_4}\\
&+\frac{\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,u_4}-\frac{i\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,|Cr-1|\sin(\theta)u_4}-\frac{\sqrt{2}C^3\frac{\partial}{\partial\theta}u_4}{4\,|Cr-1|u_4}-\frac{\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,|Cr-1|u_4^2}-\frac{i\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,\sin(\theta)u_4^2}+\frac{i\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,|Cr-1|\sin(\theta)u_4}-\frac{\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,|Cr-1|u_4}-\frac{i\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,\sin(\theta)u_4}-\frac{\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{4\,ru_4}+\frac{i\sqrt{2}C^3\frac{\partial}{\partial\phi}u_4}{4\,|Cr-1|\sin(\theta)u_4}\\
&+\frac{i\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,|Cr-1|\sin(\theta)u_4^2}-\frac{\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,|Cr-1|u_4}+\frac{i\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,|Cr-1|\sin(\theta)u_4}+\frac{\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{4\,r|Cr-1|u_4}-\frac{\sqrt{2}C\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,|Cr-1|u_4^2}+\frac{i\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{4\,r\sin(\theta)u_4}-\frac{\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,ru_4}+\frac{i\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,|Cr-1|\sin(\theta)u_4}-\frac{5\sqrt{2}C\frac{\partial^2}{\partial r\partial\theta}u_4}{24\,r|Cr-1|u_4}+\frac{\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{48\,r|Cr-1|u_4^2}\\
&-\frac{\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,ru_4^2}-\frac{i\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)u_4}+\frac{i\sqrt{2}C\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,|Cr-1|\sin(\theta)u_4^2}+\frac{\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,r|Cr-1|u_4}+\frac{i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,r\sin(\theta)u_4}-\frac{5\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{24\,r^2u_4}+\frac{5i\sqrt{2}C\frac{\partial^2}{\partial r\partial\phi}u_4}{24\,r|Cr-1|\sin(\theta)u_4}+\frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4}{4\,r^2|Cr-1|u_4}+\frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{48\,r^2u_4^2}+\frac{\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,r|Cr-1|u_4^2}\\
&+\frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{24\,r^2|Cr-1|\sin(\theta)u_4^2}+\frac{\sqrt{2}C\cos(\theta)\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^2|Cr-1|\sin(\theta)u_4}-\frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^2|Cr-1|u_4^2}+\frac{\sqrt{2}C\frac{\partial^3}{(\partial\theta)^3}u_4}{48\,r^2|Cr-1|u_4}-\frac{i\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{48\,r|Cr-1|\sin(\theta)u_4^2}+\frac{i\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,r\sin(\theta)u_4^2}-\frac{i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,r|Cr-1|\sin(\theta)u_4}+\frac{5\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{24\,r^2|Cr-1|u_4}+\frac{5i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{24\,r^2\sin(\theta)u_4}\\
&+\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,r^3u_4}-\frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{48\,r^2|Cr-1|u_4^2}+\frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{24\,r^3\sin(\theta)u_4^2}+\frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^3\sin(\theta)u_4}-\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^3u_4^2}+\frac{\sqrt{2}\frac{\partial^3}{(\partial\theta)^3}u_4}{48\,r^3u_4}-\frac{i\sqrt{2}C\frac{\partial^3}{(\partial\theta)^2\partial\phi}u_4}{48\,r^2|Cr-1|\sin(\theta)u_4}-\frac{i\sqrt{2}C\cos(\theta)\frac{\partial^2}{\partial\theta\partial\phi}u_4}{48\,r^2|Cr-1|\sin(\theta)^2u_4}+\frac{i\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2}\\
&-\frac{i\sqrt{2}C\frac{\partial}{\partial\phi}u_4}{4\,r^2|Cr-1|\sin(\theta)u_4}-\frac{i\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{48\,r^2\sin(\theta)u_4^2}-\frac{i\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,r|Cr-1|\sin(\theta)u_4^2}-\frac{i\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^2|Cr-1|\sin(\theta)^2u_4^2}-\frac{i\sqrt{2}C\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^2|Cr-1|\sin(\theta)u_4^2}-\frac{5i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{24\,r^2|Cr-1|\sin(\theta)u_4}-\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,r^3|Cr-1|u_4}-\frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4}{48\,r^2|Cr-1|\sin(\theta)^2u_4}\\
&-\frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{24\,r^3|Cr-1|\sin(\theta)u_4^2}-\frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^3|Cr-1|\sin(\theta)u_4}+\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^3|Cr-1|u_4^2}-\frac{\sqrt{2}\frac{\partial^3}{(\partial\theta)^3}u_4}{48\,r^3|Cr-1|u_4}-\frac{i\sqrt{2}\frac{\partial^3}{(\partial\theta)^2\partial\phi}u_4}{48\,r^3\sin(\theta)u_4}-\frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial\theta\partial\phi}u_4}{48\,r^3\sin(\theta)^2u_4}+\frac{i\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^3\sin(\theta)u_4^2}+\frac{\sqrt{2}C\frac{\partial^3}{\partial\theta(\partial\phi)^2}u_4}{48\,r^2|Cr-1|\sin(\theta)^2u_4}-\frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r^3\sin(\theta)u_4}\\
&+\frac{i\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{48\,r^2|Cr-1|\sin(\theta)u_4^2}-\frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^3\sin(\theta)^2u_4^2}-\frac{i\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^3\sin(\theta)u_4^2}-\frac{\sqrt{2}C\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2}+\frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^2|Cr-1|\sin(\theta)^3u_4^2}-\frac{\sqrt{2}C\cos(\theta)\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2|Cr-1|\sin(\theta)^3u_4}+\frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2|Cr-1|\sin(\theta)^2u_4^2}-\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{48\,r^3\sin(\theta)^2u_4}\\
&+\frac{i\sqrt{2}\frac{\partial^3}{(\partial\theta)^2\partial\phi}u_4}{48\,r^3|Cr-1|\sin(\theta)u_4}+\frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial\theta\partial\phi}u_4}{48\,r^3|Cr-1|\sin(\theta)^2u_4}-\frac{i\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)u_4^2}+\frac{\sqrt{2}\frac{\partial^3}{\partial\theta(\partial\phi)^2}u_4}{48\,r^3\sin(\theta)^2u_4}+\frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r^3|Cr-1|\sin(\theta)u_4}+\frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^3|Cr-1|\sin(\theta)^2u_4^2}+\frac{i\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^3|Cr-1|\sin(\theta)u_4^2}-\frac{\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3\sin(\theta)^2u_4^2}\\
&+\frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^3\sin(\theta)^3u_4^2}-\frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^3\sin(\theta)^3u_4}+\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^3\sin(\theta)^2u_4^2}+\frac{i\sqrt{2}C\frac{\partial}{\partial\phi}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,r^2|Cr-1|\sin(\theta)^3u_4^2}-\frac{i\sqrt{2}C\frac{\partial^3}{(\partial\phi)^3}u_4}{48\,r^2|Cr-1|\sin(\theta)^3u_4}+\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{48\,r^3|Cr-1|\sin(\theta)^2u_4}-\frac{\sqrt{2}\frac{\partial^3}{\partial\theta(\partial\phi)^2}u_4}{48\,r^3|Cr-1|\sin(\theta)^2u_4}+\frac{\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)^2u_4^2}\\
&-\frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^3|Cr-1|\sin(\theta)^3u_4^2}+\frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\phi)^2}u_4}{24\,r^3|Cr-1|\sin(\theta)^3u_4}-\frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^3|Cr-1|\sin(\theta)^2u_4^2}+\frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,r^3\sin(\theta)^3u_4^2}-\frac{i\sqrt{2}\frac{\partial^3}{(\partial\phi)^3}u_4}{48\,r^3\sin(\theta)^3u_4}-\frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,r^3|Cr-1|\sin(\theta)^3u_4^2}+\frac{i\sqrt{2}\frac{\partial^3}{(\partial\phi)^3}u_4}{48\,r^3|Cr-1|\sin(\theta)^3u_4}+\ldots
\end{aligned}
$$

$$
\begin{aligned}
BI8 = {} & \frac{\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,|Cr-1|u_4} + \frac{\sqrt{2}C^3r^2\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,|Cr-1|u_4^2} + \frac{\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,u_4} + \frac{i\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,|Cr-1|u_4} + \frac{\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,u_4^2} + \frac{i\sqrt{2}C^3r^2\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,|Cr-1|\sin(\theta)u_4^2} - \frac{\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,|Cr-1|u_4} \\
& + \frac{i\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,\sin(\theta)u_4} + \frac{\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,u_4} + \frac{i\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}C^3\frac{\partial}{\partial\theta}u_4}{4\,|Cr-1|u_4} - \frac{\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,|Cr-1|u_4^2} + \frac{i\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,\sin(\theta)u_4^2} - \frac{i\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{4\,|Cr-1|u_4} + \frac{i\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,\sin(\theta)u_4} \\
& - \frac{\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{4\,ru_4} - \frac{i\sqrt{2}C^3\frac{\partial}{\partial\phi}u_4}{4\,|Cr-1|\sin(\theta)u_4} - \frac{i\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,|Cr-1|\sin(\theta)u_4^2} - \frac{\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,|Cr-1|u_4} - \frac{i\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{4\,|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{4\,r|Cr-1|u_4} - \frac{\sqrt{2}C\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,|Cr-1|u_4^2} - \frac{i\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{4\,r\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,ru_4} - \frac{i\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,|Cr-1|\sin(\theta)u_4} \\
& - \frac{5\sqrt{2}C\frac{\partial^2}{\partial r\partial\theta}u_4}{24\,r|Cr-1|u_4} + \frac{\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{48\,r|Cr-1|u_4^2} - \frac{\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,ru_4^2} + \frac{i\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{4\,r|Cr-1|\sin(\theta)u_4} - \frac{i\sqrt{2}C\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,|Cr-1|\sin(\theta)u_4^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{24\,r|Cr-1|u_4} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,r\sin(\theta)u_4} - \frac{5\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{24\,r^2u_4} - \frac{5i\sqrt{2}C\frac{\partial^2}{\partial r\partial\phi}u_4}{24\,r|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4}{4\,r^2|Cr-1|u_4} \\
& + \frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{48\,r^2u_4^2} + \frac{\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{48\,r|Cr-1|u_4^2} + \frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{24\,r^2|Cr-1|\sin(\theta)u_4^2} + \frac{\sqrt{2}C\cos(\theta)\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^2|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^2|Cr-1|u_4^2} + \frac{\sqrt{2}C\frac{\partial^3}{(\partial\theta)^3}u_4}{48\,r^2|Cr-1|u_4} + \frac{i\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{48\,r|Cr-1|\sin(\theta)u_4^2} - \frac{i\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,r\sin(\theta)u_4^2} + \frac{i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{24\,r|Cr-1|\sin(\theta)u_4} \\
& + \frac{5\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{24\,r^2|Cr-1|u_4} - \frac{5i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{24\,r^2\sin(\theta)u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,r^3u_4} - \frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{48\,r^2|Cr-1|u_4^2} + \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{24\,r^3\sin(\theta)u_4^2} + \frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^3\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^3u_4^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial\theta)^3}u_4}{48\,r^3u_4} + \frac{i\sqrt{2}C\frac{\partial^3}{(\partial\theta)^2\partial\phi}u_4}{48\,r^2|Cr-1|\sin(\theta)u_4} + \frac{i\sqrt{2}C\cos(\theta)\frac{\partial^2}{\partial\theta\partial\phi}u_4}{48\,r^2|Cr-1|\sin(\theta)^2u_4} \\
& - \frac{i\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2} + \frac{i\sqrt{2}C\frac{\partial}{\partial\phi}u_4}{4\,r^2|Cr-1|\sin(\theta)u_4} + \frac{i\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{48\,r^2\sin(\theta)u_4^2} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{48\,r|Cr-1|\sin(\theta)u_4^2} + \frac{i\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^2|Cr-1|\sin(\theta)^2u_4^2} + \frac{i\sqrt{2}C\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^2|Cr-1|\sin(\theta)u_4^2} + \frac{5i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{24\,r^2|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,r^3|Cr-1|u_4} \\
& - \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4}{48\,r^2|Cr-1|\sin(\theta)^2u_4} - \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{24\,r^3|Cr-1|\sin(\theta)u_4^2} - \frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^3|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{48\,r^3|Cr-1|u_4^2} - \frac{\sqrt{2}\frac{\partial^3}{(\partial\theta)^3}u_4}{48\,r^3|Cr-1|u_4} + \frac{i\sqrt{2}\frac{\partial^3}{(\partial\theta)^2\partial\phi}u_4}{48\,r^3\sin(\theta)u_4} + \frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial\theta\partial\phi}u_4}{48\,r^3\sin(\theta)^2u_4} - \frac{i\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^3\sin(\theta)u_4^2} + \frac{\sqrt{2}C\frac{\partial^3}{\partial\theta(\partial\phi)^2}u_4}{48\,r^2|Cr-1|\sin(\theta)^2u_4} \\
& + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r^3\sin(\theta)u_4} - \frac{i\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{48\,r^2|Cr-1|\sin(\theta)u_4^2} + \frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^3\sin(\theta)^2u_4^2} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^3\sin(\theta)u_4^2} - \frac{\sqrt{2}C\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)^2u_4^2} + \frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^2|Cr-1|\sin(\theta)^3u_4^2} - \frac{\sqrt{2}C\cos(\theta)\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2|Cr-1|\sin(\theta)^3u_4} + \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^2|Cr-1|\sin(\theta)^2u_4^2} \\
& - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{48\,r^3\sin(\theta)^2u_4} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial\theta)^2\partial\phi}u_4}{48\,r^3|Cr-1|\sin(\theta)u_4} - \frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial\theta\partial\phi}u_4}{48\,r^3|Cr-1|\sin(\theta)^2u_4} + \frac{i\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{\partial\theta\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)u_4^2} + \frac{\sqrt{2}\frac{\partial^3}{\partial\theta(\partial\phi)^2}u_4}{48\,r^3\sin(\theta)^2u_4} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r^3|Cr-1|\sin(\theta)u_4} - \frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^3|Cr-1|\sin(\theta)^2u_4^2} - \frac{i\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{24\,r^3|Cr-1|\sin(\theta)u_4^2} \\
& - \frac{\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3\sin(\theta)^2u_4^2} + \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^3\sin(\theta)^3u_4^2} - \frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^3\sin(\theta)^3u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^3\sin(\theta)^2u_4^2} - \frac{i\sqrt{2}C\frac{\partial}{\partial\phi}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,r^2|Cr-1|\sin(\theta)^3u_4^2} + \frac{i\sqrt{2}C\frac{\partial^3}{(\partial\phi)^3}u_4}{48\,r^2|Cr-1|\sin(\theta)^3u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{48\,r^3|Cr-1|\sin(\theta)^2u_4} - \frac{\sqrt{2}\frac{\partial^3}{\partial\theta(\partial\phi)^2}u_4}{48\,r^3|Cr-1|\sin(\theta)^2u_4} \\
& + \frac{\sqrt{2}\frac{\partial^2}{\partial\theta\partial\phi}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^3|Cr-1|\sin(\theta)^2u_4^2} - \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\phi}u_4^2}{16\,r^3|Cr-1|\sin(\theta)^3u_4^2} + \frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^3|Cr-1|\sin(\theta)^3u_4} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{24\,r^3|Cr-1|\sin(\theta)^2u_4^2} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,r^3\sin(\theta)^3u_4^2} + \frac{i\sqrt{2}\frac{\partial^3}{(\partial\phi)^3}u_4}{48\,r^3\sin(\theta)^3u_4} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{48\,r^3|Cr-1|\sin(\theta)^3u_4^2} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial\phi)^3}u_4}{48\,r^3|Cr-1|\sin(\theta)^3u_4}
\end{aligned}
$$

$$
\begin{aligned}
BI9 = {} & \frac{\sqrt{2}\sqrt{Cr+1}C^5r^8\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4^2} + \frac{3\sqrt{2}\sqrt{Cr+1}C^5r^8\frac{\partial^3}{(\partial r)^3}u_4}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4} - \frac{\sqrt{2}\sqrt{Cr+1}C^5r^7\frac{\partial}{\partial r}u_4^2}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4^2} + \frac{7\sqrt{2}\sqrt{Cr+1}C^5r^7\frac{\partial^2}{(\partial r)^2}u_4}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4} + \frac{\sqrt{2}\sqrt{Cr+1}C^4r^7\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\,(C^2r^5-r^3)u_4^2} + \frac{3\sqrt{2}\sqrt{Cr+1}C^4r^7\sqrt{|Cr-1|}\frac{\partial^3}{(\partial r)^3}u_4}{32\,(C^2r^5-r^3)u_4} \\
& - \frac{11\sqrt{2}\sqrt{Cr+1}C^5r^6\frac{\partial}{\partial r}u_4}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4} - \frac{\sqrt{2}\sqrt{Cr+1}C^4r^6\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4^2}{32\,(C^2r^5-r^3)u_4^2} + \frac{7\sqrt{2}\sqrt{Cr+1}C^4r^6\sqrt{|Cr-1|}\frac{\partial^2}{(\partial r)^2}u_4}{32\,(C^2r^5-r^3)u_4} - \frac{\sqrt{2}\sqrt{Cr+1}C^4r^7\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4^2} - \frac{3\sqrt{2}\sqrt{Cr+1}C^4r^7\frac{\partial^3}{(\partial r)^3}u_4}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4} + \frac{3\sqrt{2}\sqrt{Cr+1}C^5r^5}{8\,(C^2r^5-r^3)\sqrt{|Cr-1|}} \\
& - \frac{\sqrt{2}C^4r^7\frac{\partial^3}{\partial t(\partial r)^2}u_4}{32\,(C^2r^5-r^3)u_4^2} - \frac{11\sqrt{2}\sqrt{Cr+1}C^4r^5\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4}{32\,(C^2r^5-r^3)u_4} + \frac{\sqrt{2}\sqrt{Cr+1}C^4r^6\frac{\partial}{\partial r}u_4^2}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4^2} - \frac{7\sqrt{2}\sqrt{Cr+1}C^4r^6\frac{\partial^2}{(\partial r)^2}u_4}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4} + \frac{\sqrt{2}C^4r^7\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\,(C^2r^5-r^3)u_4^3} + \frac{3\sqrt{2}\sqrt{Cr+1}C^4r^4\sqrt{|Cr-1|}}{8\,(C^2r^5-r^3)} + \frac{\sqrt{2}C^4r^6\frac{\partial^2}{\partial t\partial r}u_4}{32\,(C^2r^5-r^3)u_4^2} \\
& - \frac{\sqrt{2}C^3r^6|Cr-1|\frac{\partial^3}{\partial t(\partial r)^2}u_4}{32\,(C^2r^5-r^3)u_4^2} + \frac{11\sqrt{2}\sqrt{Cr+1}C^4r^5\frac{\partial}{\partial r}u_4}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4} - \frac{\sqrt{2}C^4r^6\frac{\partial}{\partial t}u_4\frac{\partial}{\partial r}u_4}{32\,(C^2r^5-r^3)u_4^3} + \frac{\sqrt{2}C^3r^6|Cr-1|\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\,(C^2r^5-r^3)u_4^3} - \frac{\sqrt{2}\sqrt{Cr+1}C^3r^6\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{16\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4^2} - \frac{3\sqrt{2}\sqrt{Cr+1}C^3r^6\frac{\partial^3}{(\partial r)^3}u_4}{16\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4} - \frac{3\sqrt{2}\sqrt{Cr+1}C^4r^4}{8\,(C^2r^5-r^3)\sqrt{|Cr-1|}} \\
& + \frac{\sqrt{2}C^3r^5|Cr-1|\frac{\partial^2}{\partial t\partial r}u_4}{32\,(C^2r^5-r^3)u_4^2} - \frac{\sqrt{2}C^3r^5|Cr-1|\frac{\partial}{\partial t}u_4\frac{\partial}{\partial r}u_4}{32\,(C^2r^5-r^3)u_4^3} + \frac{3\sqrt{2}\sqrt{Cr+1}C^3r^5\frac{\partial}{\partial r}u_4^2}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4^2} - \frac{9\sqrt{2}\sqrt{Cr+1}C^3r^5\frac{\partial^2}{(\partial r)^2}u_4}{32\,(C^2r^5-r^3)\sqrt{|Cr-1|}u_4} - \frac{\sqrt{2}\sqrt{Cr+1}C^2r^5\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{16\,(C^2r^5-r^3)u_4^2} - \frac{3\sqrt{2}\sqrt{Cr+1}C^2r^5\sqrt{|Cr-1|}\frac{\partial^3}{(\partial r)^3}u_4}{16\,(C^2r^5-r^3)u_4} - \ldots
\end{aligned}
$$

$$
\begin{aligned}
BI10 = & -\frac{7\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{32\,|Cr-1|u_4} - \frac{\sqrt{2}C^3r^2\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{32\,|Cr-1|u_4^2} - \frac{7\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{32\,u_4} - \frac{7i\sqrt{2}C^3r^2\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4} - \frac{13\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\theta}u_4}{32\,|Cr-1|u_4} + \frac{\sqrt{2}C^3r\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{32\,|Cr-1|u_4^2} - \frac{\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{32\,u_4^2} - \frac{i\sqrt{2}C^3r^2\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4^2} \\
& + \frac{7\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{32\,|Cr-1|u_4} - \frac{7i\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{32\,\sin(\theta)u_4} - \frac{13\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{32\,u_4} - \frac{13i\sqrt{2}C^3r\frac{\partial^2}{\partial r\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4} + \frac{3\sqrt{2}C^3\frac{\partial}{\partial\theta}u_4}{8\,|Cr-1|u_4} + \frac{\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{32\,u_4^2} + \frac{\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{32\,|Cr-1|u_4^2} + \frac{i\sqrt{2}C^3r\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4^2} - \frac{i\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{32\,\sin(\theta)u_4^2} \\
& + \frac{7i\sqrt{2}C^2r\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4} + \frac{13\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\theta}u_4}{32\,|Cr-1|u_4} - \frac{13i\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{32\,\sin(\theta)u_4} + \frac{3\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{8\,ru_4} - \frac{\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{32\,|Cr-1|u_4^2} + \frac{3i\sqrt{2}C^3\frac{\partial}{\partial\phi}u_4}{8\,|Cr-1|\sin(\theta)u_4} + \frac{i\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{32\,\sin(\theta)u_4^2} + \frac{i\sqrt{2}C^2r\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4^2} + \frac{7\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{32\,|Cr-1|u_4} \\
& + \frac{13i\sqrt{2}C^2\frac{\partial^2}{\partial r\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4} - \frac{3\sqrt{2}C^2\frac{\partial}{\partial\theta}u_4}{8\,r|Cr-1|u_4} + \frac{\sqrt{2}C\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{32\,|Cr-1|u_4^2} + \frac{3i\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{8\,r\sin(\theta)u_4} - \frac{i\sqrt{2}C^2\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4^2} + \frac{7\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{32\,ru_4} + \frac{7i\sqrt{2}C\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4} + \frac{5\sqrt{2}C\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,r|Cr-1|u_4} - \frac{\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r|Cr-1|u_4^2} + \frac{\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{32\,ru_4^2} \\
& - \frac{3i\sqrt{2}C^2\frac{\partial}{\partial\phi}u_4}{8\,r|Cr-1|\sin(\theta)u_4} + \frac{i\sqrt{2}C\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{32\,|Cr-1|\sin(\theta)u_4^2} - \frac{7\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\theta}u_4}{32\,r|Cr-1|u_4} + \frac{7i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{32\,r\sin(\theta)u_4} + \frac{5\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,r^2u_4} + \frac{5i\sqrt{2}C\frac{\partial^2}{\partial r\partial\phi}u_4}{16\,r|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4}{4\,r^2|Cr-1|u_4} - \frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2u_4^2} - \frac{\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\theta}u_4}{32\,r|Cr-1|u_4^2} - \frac{\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{32\,r^2|Cr-1|\sin(\theta)u_4^2} \\
& + \frac{\sqrt{2}C\cos(\theta)\frac{\partial^2}{(\partial\theta)^2}u_4}{32\,r^2|Cr-1|\sin(\theta)u_4} - \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{32\,r^2|Cr-1|u_4^2} + \frac{\sqrt{2}C\frac{\partial^3}{(\partial\theta)^3}u_4}{32\,r^2|Cr-1|u_4} - \frac{i\sqrt{2}C\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,r|Cr-1|\sin(\theta)u_4^2} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{32\,r\sin(\theta)u_4^2} - \frac{7i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}u_4}{32\,r|Cr-1|\sin(\theta)u_4} - \frac{5\sqrt{2}\frac{\partial^2}{\partial r\partial\theta}u_4}{16\,r^2|Cr-1|u_4} + \frac{5i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{16\,r^2\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,r^3u_4} + \frac{\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{16\,r^2|Cr-1|u_4^2} \\
& - \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{32\,r^3\sin(\theta)u_4^2} + \frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\theta)^2}u_4}{32\,r^3\sin(\theta)u_4} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{32\,r^3u_4^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial\theta)^3}u_4}{32\,r^3u_4} + \frac{i\sqrt{2}C\frac{\partial^3}{(\partial\theta)^2\partial\phi}u_4}{32\,r^2|Cr-1|\sin(\theta)u_4} + \frac{i\sqrt{2}C\cos(\theta)\frac{\partial^2}{\partial\theta\partial\phi}u_4}{32\,r^2|Cr-1|\sin(\theta)^2u_4} - \frac{i\sqrt{2}C\frac{\partial}{\partial\phi}u_4}{4\,r^2|Cr-1|\sin(\theta)u_4} - \frac{i\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2\sin(\theta)u_4^2} - \frac{i\sqrt{2}\frac{\partial^2}{(\partial r)^2}u_4\frac{\partial}{\partial\phi}u_4}{32\,r|Cr-1|\sin(\theta)u_4^2} \\
& - \frac{i\sqrt{2}C\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{32\,r^2|Cr-1|\sin(\theta)^2u_4^2} - \frac{i\sqrt{2}C\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{32\,r^2|Cr-1|\sin(\theta)u_4^2} - \frac{5i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{4\,r^3|Cr-1|u_4} - \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4}{32\,r^2|Cr-1|\sin(\theta)^2u_4} + \frac{\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4^2}{32\,r^3|Cr-1|\sin(\theta)u_4^2} - \frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\theta)^2}u_4}{32\,r^3|Cr-1|\sin(\theta)u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{32\,r^3|Cr-1|u_4^2} \\
& - \frac{\sqrt{2}\frac{\partial^3}{(\partial\theta)^3}u_4}{32\,r^3|Cr-1|u_4} + \frac{i\sqrt{2}\frac{\partial^3}{(\partial\theta)^2\partial\phi}u_4}{32\,r^3\sin(\theta)u_4} + \frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial\theta\partial\phi}u_4}{32\,r^3\sin(\theta)^2u_4} + \frac{\sqrt{2}C\frac{\partial^3}{\partial\theta(\partial\phi)^2}u_4}{32\,r^2|Cr-1|\sin(\theta)^2u_4} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r^3\sin(\theta)u_4} + \frac{i\sqrt{2}\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\phi}u_4}{16\,r^2|Cr-1|\sin(\theta)u_4^2} - \frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{32\,r^3\sin(\theta)^2u_4^2} - \frac{i\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{32\,r^3\sin(\theta)u_4^2} - \frac{\sqrt{2}C\cos(\theta)\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^2|Cr-1|\sin(\theta)^3u_4} \\
& - \frac{\sqrt{2}C\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{32\,r^2|Cr-1|\sin(\theta)^2u_4^2} - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{32\,r^3\sin(\theta)^2u_4} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial\theta)^2\partial\phi}u_4}{32\,r^3|Cr-1|\sin(\theta)u_4} - \frac{i\sqrt{2}\cos(\theta)\frac{\partial^2}{\partial\theta\partial\phi}u_4}{32\,r^3|Cr-1|\sin(\theta)^2u_4} + \frac{\sqrt{2}\frac{\partial^3}{\partial\theta(\partial\phi)^2}u_4}{32\,r^3\sin(\theta)^2u_4} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4}{4\,r^3|Cr-1|\sin(\theta)u_4} + \frac{i\sqrt{2}\cos(\theta)\frac{\partial}{\partial\theta}u_4\frac{\partial}{\partial\phi}u_4}{32\,r^3|Cr-1|\sin(\theta)^2u_4^2} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial\theta)^2}u_4\frac{\partial}{\partial\phi}u_4}{32\,r^3|Cr-1|\sin(\theta)u_4^2} - \frac{\sqrt{2}C\cos(\theta)\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3\sin(\theta)^3u_4} \\
& - \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{32\,r^3\sin(\theta)^2u_4^2} - \frac{i\sqrt{2}C\frac{\partial}{\partial\phi}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{32\,r^2|Cr-1|\sin(\theta)^3u_4^2} + \frac{i\sqrt{2}C\frac{\partial^3}{(\partial\phi)^3}u_4}{32\,r^2|Cr-1|\sin(\theta)^3u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4}{32\,r^3|Cr-1|\sin(\theta)^2u_4} - \frac{\sqrt{2}\frac{\partial^3}{\partial\theta(\partial\phi)^2}u_4}{32\,r^3|Cr-1|\sin(\theta)^2u_4} + \frac{\sqrt{2}\cos(\theta)\frac{\partial^2}{(\partial\phi)^2}u_4}{16\,r^3|Cr-1|\sin(\theta)^3u_4} + \frac{\sqrt{2}\frac{\partial}{\partial\theta}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{32\,r^3|Cr-1|\sin(\theta)^2u_4^2} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}u_4\frac{\partial^2}{(\partial\phi)^2}u_4}{32\,r^3\sin(\theta)^3u_4^2} + \ldots
\end{aligned}
$$

$$
\begin{aligned}
BI11 = &-\frac{\sqrt{2}C^5r^8\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}-\frac{3\sqrt{2}C^5r^8\frac{\partial^3}{(\partial r)^3}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}+\frac{\sqrt{2}C^5r^7\frac{\partial}{\partial r}u_4^2}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}-\frac{7\sqrt{2}C^5r^7\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}-\frac{\sqrt{2}C^4r^7\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4^2}\\
&-\frac{3\sqrt{2}C^4r^7\sqrt{|Cr-1|}\frac{\partial^3}{(\partial r)^3}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4}+\frac{11\sqrt{2}C^5r^6\frac{\partial}{\partial r}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}+\frac{\sqrt{2}C^4r^6\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4^2}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4^2}-\frac{7\sqrt{2}C^4r^6\sqrt{|Cr-1|}\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4}+\frac{\sqrt{2}C^4r^7\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}+\frac{3\sqrt{2}C^4r^7\frac{\partial^3}{(\partial r)^3}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}\\
&-\frac{3\sqrt{2}C^5r^5}{8\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}}+\frac{11\sqrt{2}C^4r^5\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4}-\frac{\sqrt{2}C^4r^6\frac{\partial}{\partial r}u_4^2}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}+\frac{7\sqrt{2}C^4r^6\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}-\frac{3\sqrt{2}C^4r^4\sqrt{|Cr-1|}}{8\left(Cr^4-r^3\right)\sqrt{Cr+1}}-\frac{\sqrt{2}C^3r^6\frac{\partial^3}{\partial t(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)u_4^2}\\
&-\frac{11\sqrt{2}C^4r^5\frac{\partial}{\partial r}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}+\frac{\sqrt{2}C^3r^6\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)u_4^3}+\frac{\sqrt{2}C^3r^6\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}+\frac{3\sqrt{2}C^3r^6\frac{\partial^3}{(\partial r)^3}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}+\frac{3\sqrt{2}C^4r^4}{8\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}}+\frac{\sqrt{2}C^3r^5\frac{\partial^2}{\partial t\partial r}u_4}{32\left(Cr^4-r^3\right)u_4^2}\\
&-\frac{\sqrt{2}C^2r^5|Cr-1|\frac{\partial^3}{\partial t(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)u_4^2}-\frac{\sqrt{2}C^3r^5\frac{\partial}{\partial t}u_4\frac{\partial}{\partial r}u_4}{32\left(Cr^4-r^3\right)u_4^3}-\frac{3\sqrt{2}C^3r^5\frac{\partial}{\partial r}u_4^2}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}+\frac{9\sqrt{2}C^3r^5\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}+\frac{\sqrt{2}C^2r^5|Cr-1|\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)u_4^3}+\frac{\sqrt{2}C^2r^5\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4^2}\\
&+\frac{3\sqrt{2}C^2r^5\sqrt{|Cr-1|}\frac{\partial^3}{(\partial r)^3}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4}+\frac{\sqrt{2}C^2r^4|Cr-1|\frac{\partial^2}{\partial t\partial r}u_4}{32\left(Cr^4-r^3\right)u_4^2}+\frac{\sqrt{2}C^2r^5\frac{\partial^3}{\partial t(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)u_4^2}-\frac{17\sqrt{2}C^3r^4\frac{\partial}{\partial r}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}-\frac{\sqrt{2}C^2r^4|Cr-1|\frac{\partial}{\partial t}u_4\frac{\partial}{\partial r}u_4}{32\left(Cr^4-r^3\right)u_4^3}-\frac{3\sqrt{2}C^2r^4\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4^2}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4^2}\\
&+\frac{9\sqrt{2}C^2r^4\sqrt{|Cr-1|}\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4}-\frac{\sqrt{2}C^2r^5\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)u_4^3}-\frac{\sqrt{2}C^2r^5\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}-\frac{3\sqrt{2}C^2r^5\frac{\partial^3}{(\partial r)^3}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}-\frac{3\sqrt{2}C^3r^4\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}\sin(\theta)u_4}\\
&-\frac{3\sqrt{2}C^3r^4\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}-\frac{\sqrt{2}C^3r^4\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}\sin(\theta)u_4^2}-\frac{\sqrt{2}C^3r^4\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}+\frac{\sqrt{2}C^3r^3}{4\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}}-\frac{\sqrt{2}C^2r^4\frac{\partial^2}{\partial t\partial r}u_4}{32\left(Cr^4-r^3\right)u_4^2}\\
&-\frac{17\sqrt{2}C^2r^3\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4}+\frac{\sqrt{2}C^2r^4\frac{\partial}{\partial t}u_4\frac{\partial}{\partial r}u_4}{32\left(Cr^4-r^3\right)u_4^3}+\frac{3\sqrt{2}C^2r^4\frac{\partial}{\partial r}u_4^2}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}-\frac{9\sqrt{2}C^2r^4\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}-\frac{3\sqrt{2}C^2r^3\sqrt{|Cr-1|}\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sin(\theta)u_4}-\frac{3\sqrt{2}C^2r^3\sqrt{|Cr-1|}\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4}\\
&-\frac{\sqrt{2}C^3r^3\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}\sin(\theta)u_4}-\frac{\sqrt{2}C^2r^3\sqrt{|Cr-1|}\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sin(\theta)u_4^2}-\frac{\sqrt{2}C^3r^3\frac{\partial^2}{(\partial\theta)^2}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}-\frac{\sqrt{2}C^2r^3\sqrt{|Cr-1|}\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial\theta)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4^2}+\frac{\sqrt{2}C^2r^2\sqrt{|Cr-1|}}{4\left(Cr^4-r^3\right)\sqrt{Cr+1}}+\frac{\sqrt{2}Cr^4\frac{\partial^3}{\partial t(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)u_4^2}\\
&+\frac{17\sqrt{2}C^2r^3\frac{\partial}{\partial r}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}-\frac{\sqrt{2}Cr^4\frac{\partial}{\partial t}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)u_4^3}-\frac{\sqrt{2}Cr^4\frac{\partial}{\partial r}u_4\frac{\partial^2}{(\partial r)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4^2}-\frac{3\sqrt{2}Cr^4\frac{\partial^3}{(\partial r)^3}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}+\frac{3\sqrt{2}C^2r^3\cos(\theta)\frac{\partial^2}{\partial r\partial\theta}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}\sin(\theta)u_4}\\
&+\frac{3\sqrt{2}C^2r^3\frac{\partial^3}{\partial r(\partial\theta)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}u_4}-\frac{3\sqrt{2}C^3r^4\frac{\partial^3}{\partial r(\partial\phi)^2}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}\sin(\theta)^2u_4}-\frac{\sqrt{2}C^2r^2\sqrt{|Cr-1|}\cos(\theta)\frac{\partial}{\partial\theta}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}\sin(\theta)u_4}+\frac{\sqrt{2}C^2r^3\cos(\theta)\frac{\partial}{\partial r}u_4\frac{\partial}{\partial\theta}u_4}{32\left(Cr^4-r^3\right)\sqrt{Cr+1}\sqrt{|Cr-1|}\sin(\theta)u_4^2}-\frac{\sqrt{2}C^2r^2\sqrt{|Cr-1|}\frac{\partial^2}{(\partial\theta)^2}u_4}{16\left(Cr^4-r^3\right)\sqrt{Cr+1}u_4}+\ldots
\end{aligned}
$$

Petrov Type

Type "I"